

DriveXR - A Meta Quest Link-Based Extended Reality Architecture for Physical Wheel Integration and Hand-Tracked Interaction in Driver Training Simulation

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Project report submitted in partial fulfilment of the requirements
for the Degree of M.Tech. in Computer Engineering (Specialized in Software Engineering)
on July 11, 2026

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Student Declaration

I hereby declare that the work presented in the report entitled "**DriveXR - A Meta Quest Link-Based Extended Reality Architecture for Physical Wheel Integration and Hand-Tracked Interaction in Driver Training Simulation**" submitted by me for the partial fulfilment of the requirements for the degree of *M.Tech. in Computer Engineering(Specialized in Software Engineering)* at Veermata Jijabai Technological Institute, Mumbai, is an authentic record of my work carried out under the guidance of **Dr. V.B.Nikam** . Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other degree.

Ritvik Babre

Place & Date: July 11, 2026

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dr. V.B.Nikam

Place & Date: July 11, 2026

Project Report Approval

This is to certify that the project entitled

**“DriveXR – A Meta Quest Link-Based Extended Reality
Architecture for Physical Wheel Integration and Hand-Tracked
Interaction in Driver Training Simulation”**

is approved in completion of Project Dissertation for the degree of M.Tech
in Computer Engineering (specialized in Software Engineering).

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Date: ____/____/2026

CERTIFICATE

This is to certify that work entitled “**DriveXR – A Meta Quest Link-Based Extended Reality Architecture for Physical Wheel Integration and Hand-Tracked Interaction in Driver Training Simulation**” is the work of Annual Progress Stage – IV of the Computer Engineering Branch, Academic Year 2025–2026 and Master of Technology Program by **Ritvik Babre (ID: 242190002)**. He has satisfactorily completed the required assignment for the Second Year 2025–26 as laid down by the Institute.

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Contents

Abstract

Traditional driver training methods face significant limitations in terms of safety, cost, and scalability, particularly when training on high-end vehicles or in hazardous scenarios. This project presents the design and implementation of DriveXR, a Virtual Reality (VR) driving simulator built to help novice drivers safely practise vehicle control before driving on public roads. The system is developed in Unity 6 and runs as a PC-hosted Windows application, streaming visual output to a Meta Quest 3 headset via Meta Quest Link. This architecture retains full six-degrees-of-freedom head tracking and optical hand tracking from the Quest 3's onboard cameras while leveraging desktop-class GPU rendering, avoiding the platform limitations of a fully standalone deployment. Physical input is provided by a Logitech G29 racing wheel and pedal set connected directly to the PC via USB, offering realistic steering, throttle, and braking control substantially more immersive than gamepad or keyboard input. A Tata Safari 2024 is used as the demonstration vehicle, modelled with Unity WheelCollider-based physics tuned to approximate real-world handling characteristics. The simulator provides three structured training environments of progressive difficulty: a straight bounded highway for control familiarisation, a winding road for handling practice, and a signalised city road grid populated with AI-controlled traffic for advanced scenario training. The NPC traffic system uses a state-machine-based controller incorporating obstacle avoidance, traffic light compliance, and intersection yielding behaviour. VR hand tracking enables direct in-cockpit interaction with the steering wheel and horn. Preliminary testing with the supervising faculty and additional participants identified key usability factors including simulator sickness adaptation patterns in VR-inexperienced users, the influence of physical hardware ergonomics on control legibility, and headset height calibration requirements. This report presents the literature review, system architecture, implementation details, and preliminary evaluation results that document the development and validation of DriveXR as an accessible, affordable driving practice platform.

Keywords: Virtual Reality, Driving Simulator, Driver Training, Meta Quest 3, Meta Quest Link, Unity 6, Logitech G29, NPC Traffic Simulation, Hand Tracking

Chapter 1

Introduction

1.1 Background and Motivation

The integration of Virtual Reality (VR) technologies in driver training has emerged as a promising approach to creating realistic and safe learning environments. Traditional driving instruction faces persistent limitations including safety concerns during hazardous scenario training, prohibitive vehicle costs for diverse fleet exposure, environmental constraints limiting practice opportunities, and scalability challenges for growing driving school enrollment. These limitations are particularly acute when training involves higher-end vehicles, where operational costs, insurance premiums, and maintenance expenses render traditional hands-on training economically impractical for many driving schools and individual learners [Harari2021].

Recent advancements in consumer VR hardware, particularly standalone headsets such as the Meta Quest 3, have reached performance and affordability thresholds enabling deployment in educational contexts. Modern VR systems deliver stereoscopic rendering at high refresh rates with low motion-to-photon latency, wide fields of view, and inside-out tracking eliminating external sensor requirements. The Quest 3 additionally offers native optical hand tracking and, critically for this project, support for Meta Quest Link — a feature enabling the headset to operate as a PC-VR display and tracking peripheral while a Unity application runs natively on a connected Windows PC.

This convergence of technologies creates an opportunity to develop immersive driving simulations that closely replicate real-world vehicle dynamics, visual fidelity, and driver control feedback using consumer-accessible hardware. This project presents **DriveXR**, a VR-based driving simulator combining a Meta Quest 3 headset with a Logitech G29 racing wheel and pedal set, running as a PC-hosted application via Meta Quest Link. The system is intended to help novice drivers safely practise fundamental vehicle control skills before driving on public roads.

1.2 Problem Statement

Despite growing interest in VR driving simulation, current systems exhibit limitations that hinder educational adoption:

1. **High Cost Barriers:** Commercial driving simulators used in some professional training contexts often cost tens of thousands of dollars per unit, placing them beyond the financial reach of most driving schools and educational institutions.
2. **Platform-Hardware Incompatibility:** Physically realistic racing wheel hardware communicates over USB HID protocols requiring host-side driver support. This creates a direct incompatibility with fully standalone VR headset deployment, a constraint not always addressed in existing lightweight VR driving research platforms.
3. **Entertainment vs Education Focus:** Commercially available racing simulator software targets entertainment use cases rather than structured driver education, generally lacking progressive-difficulty training environments or NPC traffic behaviour suited to intersection navigation practice.
4. **Limited Hardware Fidelity Evaluation:** Few accessible projects document the practical usability trade-offs between budget and premium physical control hardware specifically in the context of VR-headset-worn use, where users cannot visually reference the pedals or controls they are operating.
5. **Limited Validation:** Few studies demonstrate empirical transfer of VR training to real-world driving performance, creating uncertainty about pedagogical effectiveness for driving schools considering adoption.

There exists a clear opportunity for an accessible, consumer-hardware-based VR driving simulator that resolves the platform-hardware compatibility constraint, supports structured progressive-difficulty training environments, and documents the practical engineering and usability trade-offs involved — informing both this project’s own development and future similar systems.

1.3 Research Objectives

The primary objective of this project is to design, implement, and preliminarily evaluate a VR driving simulator built from consumer-accessible hardware, providing an immersive and realistic training experience suitable for novice driver skill development. The specific objectives are:

1. **System Architecture Design:** Develop a VR simulation architecture integrating a Meta Quest 3 headset, Unity 6 game engine, and commercial racing wheel hardware, resolving the platform compatibility constraints identified during development.

2. **Realistic Vehicle Simulation:** Implement physics-based vehicle dynamics for a Tata Safari model, tuned to approximate real-world handling characteristics using Unity’s WheelCollider physics system.
3. **Physical Control Integration:** Integrate a Logitech G29 racing wheel and pedal set with correct axis mapping and ergonomic pedal configuration, informed by direct comparison against a budget alternative.
4. **VR Hand Interaction:** Implement hand-tracking-based interaction for in-cockpit controls including the steering wheel and horn, alongside physical hardware input.
5. **Training Environment Development:** Build a progressive-difficulty set of training environments — straight highway, curved road, and signalised city grid — with an adaptive NPC traffic system for the city environment.
6. **Preliminary Usability Evaluation:** Conduct informal testing sessions to identify key usability factors including simulator sickness patterns, hardware ergonomics, and VR headset calibration requirements, informing a future formal user study.

1.4 Scope and Limitations

1.4.1 In Scope

The system development encompasses:

- A PC-hosted VR application streamed to the Meta Quest 3 via Meta Quest Link.
- Simulation of a single vehicle model (Tata Safari) with tuned WheelCollider-based physics.
- Three training environments of progressive difficulty: straight highway, curved road, and signalised city grid with NPC traffic.
- Automatic transmission simulation (Park, Reverse, Neutral, Drive).
- Physical racing wheel and pedal input via Logitech G29, with gear shifter support via Logitech Driving Force Shifter.
- VR hand-tracking interaction for the steering wheel and horn.
- Informal preliminary testing with the supervising faculty member and additional volunteer participants.

1.4.2 Out of Scope

The following are explicitly excluded from the current project phase:

- Force feedback steering resistance (hardware-capable but not yet integrated in software).
- Manual transmission / clutch-engagement simulation.
- Motion platform or vestibular simulation.
- Multi-user networked scenarios.
- Automated performance scoring and assessment logging.
- Formal regulatory certification or driving-school pilot deployment.
- Formal structured user study with standardised instruments (SSQ, SUS) — identified as near-term future work.

1.5 Research Organization

The remainder of this document is organised as follows:

Chapter ?? (Literature Review): Presents a synthesis of research papers on VR driving simulation, autonomous vehicle training, physiological monitoring, and simulator design, followed by a systematic gap analysis.

Chapter ?? (System Analysis and Design): Details functional and non-functional requirements, design alternatives considered for VR platform, game engine, and control input architecture, and the resulting system architecture.

Chapter ?? (Methodology): Documents the development methodology as executed, including the architectural pivot from a proposed standalone/custom-hardware approach to the implemented PC-hosted Meta Quest Link architecture with commercial racing wheel hardware, and the phases of development actually carried out.

Chapter ?? (Implementation): Presents the concrete implementation of DriveXR’s principal subsystems — vehicle physics, hardware input, VR and XR interaction, NPC traffic, mirror rendering, and environment construction — illustrated with screenshots and representative code excerpts.

Chapter ?? (Testing): Documents the functional testing carried out during development as a set of structured test cases, covering vehicle control, VR and XR interaction, and NPC traffic defects identified and resolved.

Chapter ?? (Results): Presents preliminary evaluation findings from informal testing, covering hardware fidelity, VR comfort, NPC traffic behaviour, and system performance.

Chapter ?? (Conclusion and Future Work): Summarises the project’s contributions, acknowledges current limitations, and outlines a roadmap for continued development.

This structured presentation documents both the academic grounding and the practical engineering process behind DriveXR’s development, including the concrete technical constraints that shaped its final architecture.

Chapter 2

Literature Review

Textual Synthesis

The collected body of work demonstrates that virtual reality driving research has progressed from early feasibility demonstrations to targeted investigations of specific sensory, cognitive, and interaction factors that shape training effectiveness and system validity. A central and repeated finding is that *interaction fidelity* – the presence of realistic physical control hardware (steering wheel, pedals, gear interface, sometimes motion cues) – exerts a stronger influence on motor skill transfer, takeover quality, and trust in automation than visual display fidelity alone [Harari2021, He2022]. High-fidelity setups accelerate embodiment of control dynamics, shorten reaction windows during automated mode disengagement, and raise subjective confidence compared to low-fidelity keyboard or abstract controller conditions.

Gamification and purposeful in-vehicle task design emerge as viable strategies for sustaining vigilance in both autonomous passenger contexts and supervised automation scenarios. Studies using gamified AR/VR overlays show that structured secondary tasks can maintain hazard detection performance while producing interpretable performance curves (learning, plateau, decline) that indirectly encode driver cognitive state [Muguro2021, Muguro2023]. This supports a shift toward multi-modal engagement models in which task metrics, gaze allocation, and interaction cadence supplement traditional camera-based monitoring. Physiological sensing (electrodermal activity, pupil dilation, heart rate variability) further reinforces these behavioral indicators by offering objective correlates of arousal, workload, and sustained attention [Qadir2019, Muguro2023, Abdurrahman2021]. Together, layered behavioral and physiological monitoring point to an emerging design space for adaptive difficulty and proactive safety interventions.

Visual perceptual factors remain salient. Dynamic Visual Acuity (DVA) has been shown to outperform static acuity in predicting unsafe lane keeping and navigation variability in VR driving tasks [Hwang2020]. This indicates that training pipelines which incorporate dynamic visual tracking exercises may more effectively identify learners requiring targeted remediation. At the same time, motion and simulator sickness mitigation remains incomplete: alternative locomotion paradigms (e.g., teleportation) reduce discomfort [Lindal2019], yet long-duration curriculum studies and sys-

tematic countermeasure evaluations (field of view tuning, vestibular cue alignment, scheduled breaks) are still scarce. Physiological motion discomfort indicators (e.g., gastric rhythms) provide promising objective metrics for evaluating mitigation strategies [Trofimova2021].

Demographic and individual difference analyses show performance stratification by age, gender, and personality traits [Chang2020, Ju2024]. Older or inexperienced users adapt more slowly to novel interfaces, while certain personality dimensions correlate negatively with perceived realism. These findings underscore the need for inclusive hardware ergonomics (adjustable force feedback profiles, simplified haptic layouts) and personalization layers that optimize challenge, feedback modality, and scenario pacing per learner profile. Eye-tracking based entropy measures further discriminate novice from experienced gaze behavior [Chung2022], suggesting integration of attention diagnostics into formative assessment rubrics.

Multi-sensory immersion work demonstrates additive benefits of synchronized vibration, motion platforms, and advanced spatial audio for perceived immersion and situational awareness [GilCarvajal2024, Hartfiel2019]. However, cost and engineering complexity still constrain widespread adoption in educational contexts. Open-source and modular platforms such as DReyeVR provide an intermediate fidelity path, enabling research-grade logging (gaze, head pose, vehicle kinetics) at significantly lower capital expenditure [Silvera2022]. Similarly, VR-assisted dataset generation approaches leverage human-in-the-loop driving to create richly annotated perception corpora for autonomous vehicle model training [Yao2020], accelerating scenario diversity while lowering collection risk.

Across studies a persistent methodological gap is the lack of *standardized assessment frameworks*. Metrics vary widely (takeover time, lane deviation, gaze entropy, cognitive load class accuracy, posture change, sickness indices), complicating cross-study synthesis and regulatory dialogue. Very few works pursue longitudinal retention or real-world transfer validation beyond simulator endpoints [He2022]. Sample sizes frequently remain modest (often under 70 participants) and demographically narrow [Muguro2023, Qadir2019, Schultheis2005], limiting external validity. Trust dynamics between human drivers and autonomous systems highlight intervention behavior and explainability gaps: participants often intervene due to insufficient model transparency or perceived scenario novelty [Xu2022].

Emerging research directions therefore converge on: (1) cost-effective, wireless, dual-mode (manual and automatic) high-fidelity control assemblies to broaden institutional adoption; (2) adaptive, physiology-informed training loops that adjust scenario complexity in real time [Abdurrahman2021]; (3) standardized multi-dimensional performance rubrics aligned with licensing criteria and safety benchmarks; (4) robust sickness mitigation protocols validated over multi-hour curricula; (5) inclusive, personalized interface ergonomics and accessibility features for seniors and at-risk groups; (6) scalable open-source ecosystems combining research instrumentation, dataset generation, and replay/multi-perspective analysis capabilities [Goedicke2022]; and (7) formal validation pipelines establishing comparability of VR training hours to on-road experience for regulatory certification. The synthesis underscores that while core enabling technologies (hardware fidelity, sensing, open platforms) are maturing, empirical depth (diversity, longitudinal retention, standardized metrics, regulatory pathways) remains the frontier that must be addressed to translate VR driving simula-

tion from promising augmentation to widely accredited training infrastructure.

Tabular Summary of Reviewed Studies

Table 2.1: Summary of key prior research on VR and AR systems for driving and autonomous vehicle simulation.

Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
Virtual Reality Tour for First-Time Users of Highly Automated Cars: Comparing the Effects of Virtual Environments with Different Levels of Interaction Fidelity (2021) [Harari2021]	Rayan Ebnali Harari, Richard Lamb, Razieh Fathi, Kevin Hulme	High-Fidelity VR with steering wheel/pedals improved automation trust, takeover time, and takeover quality vs. Low-Fidelity VR.	Participants expected takeover events, short 1-hour sessions, limited generalization to SAE L3/L4.	Shows that high interaction fidelity boosts motor-skill transfer in VR training for automated driving.
Feasibility of AR-VR Use in Autonomous Cars for User Engagements and its Effects on Posture and Vigilance During Transit (2023) [Muguro2023]	Joseph Muguro, Pringgo Widyo Laksono, Yuta Sasatake, Muhammad Ilhamdi Rusydi, Kojiro Matsushita, Minoru Sasaki	VR tasks kept users alert based on EDA and pupil size. Delay in hazard recognition < 1s. Mixed tasks improved posture.	Very small sample (15), limited scenarios, motion sickness not analyzed.	Physiological proof that VR/AR interactions help maintain vigilance in ADS.
User Monitoring in Autonomous Driving System Using Gamified Task: A Case for VR/AR In-Car Gaming (2021) [Muguro2021]	Joseph K. Muguro, Pringgo Widyo Laksono, Yuta Sasatake, Kojiro Matsushita, Minoru Sasaki	Gamified AR tasks didn't hurt hazard detection. Game score trends help estimate user state. Gaze data showed focused attention.	VR simulation instead of real vehicle; small student sample (13).	Supports use of VR/AR gamification to track user state and maintain attention.
Driving Performance Evaluation Correlated to Age and Visual Acuties Based on VR Technologies (2020) [Hwang2020]	Sooncheon Hwang, Sunhoon Kim, Dongmin Lee	Dynamic Visual Acuity predicted unsafe driving better than Static VA. Poor DVA = higher lane deviation.	Small sample (65), VR differs from real driving, same thresholds for SVA/DVA.	Shows VR's usefulness for testing how vision affects driving behavior.

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Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
VR-based Dataset for Autonomous-Driving System (2020) [Yao2020]	Shouwen Yao, Jiahao Zhang, Yu Wang	VR used to build a large and varied autonomous-driving dataset with auto-generated labels for 2D and 3D data. Human-in-the-loop VR driving captured stereo video, LiDAR, GPS across many environments and conditions.	Accuracy limited by reliance on VR sensor models and physics; still differs from real-world driving; realism constraints in simulation.	Helpful for generating large, consistent, low-cost datasets to train and test autonomous-driving algorithms in controlled scenarios.
The Effects of Age, Gender, and Control Device in a Virtual Reality Driving Simulation (2020) [Chang2020]	Wen-Te Chang	Age strongly influenced VR driving/navigation performance; young participants outperformed seniors. Straight (easy) routes gave higher scores than curved (difficult) ones. Young males performed better than young females; seniors showed no gender difference. Handlebar device improved performance only in young users, not seniors.	Seniors adapted slowly to VR; only two task types; limited VR realism; control devices not designed specifically for seniors.	Useful for understanding how age, gender, and input devices affect VR driving performance—important for designing VR training or in-car interfaces for diverse user groups.
Predicting Perceived Realism in Virtual Reality Driving Simulations Using Participants' Personality Traits, Heart Rate Changes, and Risk Preference (2024)	Uijong Ju, Sanghyeon Kim	Psychopathy and Machiavellianism negatively correlated with perceived realism; heart rate increases and risky decision-making were positively correlated. Tree-based ML (random forest, gradient boosting) predicted perceived realism best when using significant offline features; online-only models showed different strengths.	Sample skewed toward young men; events occurred once per participant; moderate baseline realism due to lack of motion cues and complex traffic; limited generalizability.	Demonstrates how personality and physiological metrics predict perceived realism and offers ML methods to model realism — valuable for adaptive VR training and improving validity of driving simulations.

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Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
Quantitative Analysis of Cognitive Load Test While Driving in a VR vs Non-VR Environment (2019) [Qadir2019]	Zeeshan Qadir, Eashita Chowdhury, Lidia Ghosh, Amit Konar	EEG-based cognitive load comparison showed VR driving produced higher brain activation (prefrontal, frontal, parietal) and higher cognitive load across all driving actions. Proposed CIT2FS classifier achieved higher accuracy in VR (avg 86.1%) vs non-VR (78.9%). VR improved classification of accident, steering, braking, etc.	Small participant pool (11); short 15-min sessions; limited ecological validity; tested only one VR hardware setup; no qualitative user-experience evaluation.	Shows VR induces stronger cognitive load and yields more accurate EEG-based driver-state classification, supporting VR's value for studying cognitive load and training autonomous-driving safety systems.
Comparison of Teleportation and Fixed-Track Driving in VR (2019) [Lindal2019]	Páll J. Lín-dal, Kamilla Rún Jóhannsdóttir, Unnar Kristjánsson, Nina Lensing, Anna Stühmeier, Annika Wohlan, Hannes H. Vilhjálms-son	Teleportation caused significantly less simulation sickness and maintained more positive attitudes toward VR compared to fixed-track driving. Heart-rate patterns suggested higher mental effort during teleportation, but overall comfort was better.	Focused on locomotion comfort, not driving skill; urban exploration task, not a driving-specific simulator; VR duration differences between groups influenced sickness.	Helps justify why VR training environments must carefully choose locomotion methods. Supports our claim that “comfortable VR motion” is essential before teaching driving skills in VR—reducing sickness improves training acceptance, user trust, and realism.

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Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
Driving Simulator Using Virtual Reality Tools Combining Sound, Vision, Vibration, and Motion (2024) [GilCarvajal2024]	Juan Camilo Gil-Carvajal, Eun Soo Jo, Dong Chul Park, Wookeun Song, Cheol-Ho Jeong	Multimodal VR driving simulator tested with sound (headphones vs loudspeakers), motion, and vibration. Motion + vibration significantly boosted immersion and powerfulness; motion contributed more than vibration. 64-speaker ambisonics gave best sound-localization accuracy; headphone-based audio (no head-tracking) performed worst. Sound reproduction method did not significantly change immersion/powerfulness.	Small sample sizes (10–20 per experiment); no individualized HRTFs; no head-tracking in headphone playback; motion platform may introduce unintended vibrations; realism limited by recording setup and controlled scenarios.	Strong support for using multi-sensory VR (motion + vibration + visuals) to achieve high-immersion driving experiences. Helps justify including motion cues in our project and shows why audio-localization fidelity matters for realism in VR driving simulations.
Evaluating VR Driving Simulation from a Player Experience Perspective (2017) [Walch2017]	Marcel Walch, Philipp Hock, Julian Frommel, David Dobbels, Katja Rogers, Michael Weber, Felix Schüssel	VR increased real-world dissociation and was preferred by participants, but did not significantly improve immersion, presence, enjoyment, or driving performance compared to triple-screen setups. VR caused slightly higher discomfort. Driving times were similar across both conditions.	Small sample (18 after exclusions); relied on racing-game software; limited realism (no traffic, constrained track); VR HMD had smaller FOV; results based on self-reported questionnaires; possible survey fatigue.	Useful for showing that VR is not automatically superior to flat-screen simulators. Supports our argument that VR must be designed carefully—comfort, FOV, realism, and matching physical/virtual controls matter for effective training in autonomous-driving contexts.

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Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
HMD-Based VR Tool for Traffic Psychological Examination: Conceptualization and Design Proposition (2021) [Jurik2021]	Vojtěch Juřík, Václav Linkov, Petr Dečký, Sára Klečková, Edita Chvojková	Paper proposes a VR-based system for traffic-psychological exams using HMDs + eye-tracking + haptic driving interface. Highlights that VR can measure complex driver traits like situational awareness, distraction, fatigue, and hazard perception more effectively than traditional tests. Emphasizes that VR enables controlled, repeatable scenarios and richer behavioural/physiological data.	Conceptual (not experimentally validated), no user study, no empirical performance data, proposed system not implemented, relies on future development of affordable ET-enabled HMDs.	Useful as a theoretical foundation showing why VR is suitable for driver assessment, supporting our justification for using VR in automated-driving research and training. Reinforces the value of eye-tracking, haptics, and controlled VR environments.
Evaluation of the Training Effect of the Driving Simulator + VR on Driving School Trainees (2022) [He2022]	Chenxi He, Yiping Wu, Xiaohua Zhao, Yang Ding, Shuo Liu	Study with 119 driving-school trainees showed strong improvements in reaction, knowledge, driving behaviour, and exam pass rates using a combined VR + simulator training system.	Limited to trainees; no long-term retention or real-world transfer tracking.	Strong evidence that VR + simulator fusion improves driving-school outcomes and supports using VR for driver-training effectiveness.
Analyzing the Inconsistency in Driving Patterns Between Manual and Autonomous Modes Under Complex Driving Scenarios with a VR-Enabled Simulation Platform (2022) [Xu2022]	Zheng Xu, Yihai Fang, Nan Zheng, Hai L. Vu	VR-based naturalistic platform + ML-driven AV model compared human vs. AV behavior. AV maintained stable speed and low accident rate (<1%), while humans hesitated and made unsafe maneuvers. 81% of drivers intervened due to low trust.	AV struggled in new environments until 60 iterations; TTC threshold unreliable; small sample (18); limited VR realism; only Level-4 autonomy tested.	Shows VR is effective for analyzing human-AV trust, intervention behavior, and differences between human-driven and autonomous patterns.

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Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
Electrogastrography in Autonomous Vehicles—An Objective Method for Assessment of Motion Sickness in Simulated Driving Environments (2021) [Trofimova2021]	Ekaterina Trofimova, Timo Koch, Thomas Ludwig	EKG reliably detected motion-sickness onset in VR autonomous-vehicle rides. Dynamic routes increased abnormal gastric rhythms and sweating; button-press self-reports matched physiological data. Heart rate was less useful.	Small sample; VR motion realism limits; only Level-5 scenario tested; EKG requires careful sensor placement.	Shows VR can objectively measure passenger discomfort using physiological signals—useful for evaluating comfort in AV/VR driving research.
Stopping Behavior in a VR Driving Simulator: A New Clinical Measure for the Assessment of Driving (2005) [Schultheis2005]	Maria T. Schultheis, Lisa K. Simone, Emily Roseman, Richard Nead, Jose Rebimbas, Ronald Mourant	VR simulator captured detailed stop-sign behavior differences between adults with and without acquired brain injury (ABI). ABI group showed more missed stops, longer learning curves, and different stopping distances.	Small sample (24); outdated VR hardware; focused only on stop-sign behavior; excludes participants with simulation sickness.	Demonstrates VR's ability to measure fine-grained driving performance, supporting its use for assessing impaired or at-risk drivers.
The Static and Dynamic Analyses of Drivers' Gaze Movement Using VR Driving Simulator (2022)	Jiyong Chung, Hyeokmin Lee, Hosang Moon, Eunghyuk Lee	Compared novice vs. experienced drivers using VR + eye-tracking. Novices had longer dwell times, longer fixations, narrower visual search, lower stationary gaze entropy, and simpler gaze-transition patterns. Experienced drivers distributed gaze more evenly and showed higher gaze-transition entropy, indicating richer situational awareness.	Small sample (23 male drivers); only intersection scenario tested; eye-tracking at 30 Hz; only VR-HMD; no diverse age groups or real-road validation.	Shows how VR + eye-tracking reveals differences in attention and hazard detection—useful for VR-based training, risk assessment, and identifying at-risk drivers.
Comparing VR and Non-VR Driving Simulations: An Experimental User Study (2017) [Weidner2017]	Florian Weidner, Anne Hoesch, Sandra Poeschl, Wolfgang Broll	Compared 2D, S3D, and VR-HMD using the Lane Change Task. Driving performance and physiological responses showed no major differences across display types. VR-HMD caused significantly higher simulator sickness than S3D.	Older VR hardware (Oculus DK2); simple task may not show depth-related differences; mostly young participants; limited vehicle-physics realism.	Shows VR does not automatically improve performance but can increase sickness—important for selecting appropriate VR hardware and scenario design.

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Table 2.1 – continued from previous page

Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
Effects of Neuro-Cognitive Load on Learning Transfer Using a VR-Based Driving System (2021) [Abdurrahman2021]	Usman A. Abdurrahman, Shih-Ching Yeh, Yunying Wong, Liang Wei	VR driving simulator + sensors (eye-tracking, pupil dilation, heart rate) showed cognitive load rises on complex routes. Harder routes caused larger pupil dilation, higher heart rate, and more mistakes. ML models reached 97% accuracy in classifying cognitive-load levels.	Conducted only in VR; 98 inexperienced student drivers; sensor noise and missing data; no real-world validation.	Shows VR + physiological sensing can objectively measure mental workload; useful for ADS training, difficulty analysis, and user-state monitoring.
DReyeVR: Democratizing Virtual Reality Driving Simulation for Behavioural & Interaction Research (2022) [Silvera2022]	Gustavo Silvera, Abhijat Biswas, Henny Admoni	Introduces DReyeVR, an open-source VR driving simulator (Unreal + CARLA) with eye-tracking, head-tracking, mirrors, spatial audio, custom scenarios, logging, and ROS support. Low-cost (<\$5000) behavioural research platform.	Medium-fidelity (no motion base); consumer VR hardware limits realism; Windows-only SDK; lacks full behaviour-transfer validation.	Highly relevant: enables flexible AV/driver behaviour studies, trust research, hazard-scenario creation, and attention monitoring in VR.
Driving Simulator with VR Glasses for Evaluation of New Interior Concepts (2019) [Hartfiel2019]	Bert Hartfiel, Alexander Kroys, Nico Kruithof, Rainer Stark	Volkswagen built a dynamic VR simulator combining an HMD, motion platform, and adjustable physical mock-up. Enables early interior/HMI evaluation with vestibular cues, realistic visuals, adjustable seats/dashboards, and synchronized audio/haptics.	Hard to calibrate motion compensation; HMD tracking issues on moving platform; consumer VR limits fidelity; no behavioural validation; costly engineering setup.	Shows how VR + motion + physical mock-ups support early design, ergonomics testing, visibility checks, and interior-concept validation for driving/AV research.

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Title (Year)	Authors	Key Findings	Gaps / Limitations	Relevance / Context
Freeway Traffic Safety Evaluation Using Virtual Reality: Focus on Compound Curve (2022) [Zhang2022]	Chi Zhang, Bo Wang, Yongchun Li, Lei Hou, Min Zhang, Changhe Liu, Zilong Xie	Built a VR-based driving simulation + human-computer interaction safety platform to study compound-curve freeway safety. Collected driver heart-rate changes, steering-wheel angle change rate, and driving-trajectory deviation to compute a comprehensive safety index (H). VR allowed testing multiple curve geometries using orthogonal experiments; results identified which curve-design factors significantly impact safety.	Limited to compound curves on Chinese freeway geometry; small participant pool; relies on simulator realism; physiological metrics can vary across individuals; no real-world driving validation.	Relevant because it shows how VR driving simulations can evaluate risky road geometries, analyze driver stability, and measure physiological load—supporting VR as an effective tool for traffic-safety assessment and design evaluation.
Rerun: Enabling Multi-Perspective Analysis of Driving Interaction in VR (2022) [Goedicke2022]	David Goedicke, Harald Haraldsson, Navit Klein, Lunshi Zhou, Avi Parush, Wendy Ju	Introduces Rerun, a Unity-based system that records VR driving interactions and allows full multi-perspective replay (first-person, third-person, observer view). Enables detailed post-hoc study of signalling, communication, and interaction between drivers, especially in multi-user VR driving setups.	No built-in behavioural validation study; relies on StrangeLand simulator; limited to visual replay (no physical cues); requires specific Unity asset (Ultimate Replay).	Useful for analysing driving communication, reconstructing participant interactions, and studying trust/attention in VR driving environments—supports high-detail behavioural evaluation in VR driving research.

The studies summarized in Table ?? collectively indicate that interaction fidelity, physiological engagement monitoring, and gamification significantly influence driver training outcomes and user vigilance. These insights directly inform the system design of the proposed VR-based driving simulator presented in Chapter ??.

2.1 Research Gap Analysis

Based on the comprehensive literature review, several critical research gaps have been identified in the current state of VR-based driving simulation and training systems:

2.1.1 Hardware and Physical Interaction Fidelity

Gap 1: Affordable High-Fidelity Control Systems. While studies [Harari2021, GilCarvajal2024] emphasize the importance of realistic physical controls (steering wheel, pedals, motion platforms), most high-fidelity systems remain prohibitively expensive for educational institutions and driving schools. There is limited research on developing low-cost (<\$5000), modular, and easily maintainable hardware solutions that preserve haptic realism and force feedback essential for motor skill transfer.

Gap 2: Seamless Bluetooth/Wireless Integration. Existing simulators predominantly rely on wired USB connections or proprietary hardware interfaces. The integration of consumer-grade Bluetooth-enabled control assemblies with minimal latency and reliable pairing mechanisms for VR driving training remains under-explored, particularly for standalone VR headsets like Meta Quest 3.

Gap 3: Dual-Mode Transmission Training. Most studies focus exclusively on either automatic or manual transmission vehicles. The development of a unified VR training platform that seamlessly switches between automatic and manual modes—allowing learners to experience clutch engagement, gear shifting mechanics, and corresponding vehicle dynamics—has not been adequately addressed.

2.1.2 Methodological and Sample Limitations

Gap 4: Small and Homogeneous Participant Samples. A significant majority of reviewed studies [Muguro2023, Qadir2019, Xu2022, Schultheis2005, Chung2022] suffer from small sample sizes (11–65 participants) and demographic homogeneity (predominantly young male university students). This severely limits the generalizability of findings to diverse driver populations including seniors, female learners, and drivers with varying levels of experience.

Gap 5: Longitudinal Evaluation and Real-World Transfer. Few studies [He2022] measure long-term skill retention or validate transfer of VR-acquired skills to actual on-road driving performance. Short experimental sessions (15 minutes to 1 hour) do not adequately assess whether VR training produces durable behavioral changes or improves real-world driving safety outcomes.

Gap 6: Motion Sickness and Comfort Over Extended Use. While some studies [Lindal2019, Trofimova2021, Weidner2017] acknowledge simulator sickness, comprehensive investigations into mitigating VR-induced nausea during extended training sessions (e.g., multi-hour driving school curricula) are lacking. Optimal locomotion methods, frame rates, field-of-view settings, and break schedules for prolonged VR exposure remain understudied.

2.1.3 Pedagogical and Training Effectiveness

Gap 7: Standardized Assessment Metrics for Driver Training. Existing research employs inconsistent evaluation metrics (reaction time, lane deviation, takeover quality, gaze entropy, etc.) [Harari2021, Hwang2020, Chung2022, Zhang2022]. There is no consensus on a uni-

fied, validated assessment framework specifically designed for VR-based driver training that aligns with real-world licensing exam standards or regulatory requirements (e.g., RTO testing protocols in India).

Gap 8: Adaptive and Personalized Training Pathways. Most VR driving simulators offer static scenarios with fixed difficulty levels. Research on adaptive training systems that dynamically adjust scenario complexity, traffic density, weather conditions, and hazard frequency based on individual learner performance and cognitive load [Abdurrahman2021, Ju2024] remains insufficient.

Gap 9: Integration with Driver Education Curricula. Few studies [He2022] examine how VR simulators can be systematically integrated into existing driving school curricula, including instructor training, lesson planning, regulatory compliance, and cost-benefit analysis for educational institutions.

2.1.4 Technical and Realism Constraints

Gap 10: Photorealistic Graphics and Diverse Environments. While open-source platforms like DReyeVR [Silvera2022] and datasets [Yao2020] provide flexibility, most VR driving simulations lack photorealistic graphics, diverse environmental conditions (urban, rural, highway, adverse weather), and accurate vehicle-specific interior models (e.g., luxury car dashboards, control layouts).

Gap 11: Multi-Sensory Feedback Beyond Vision and Sound. Although [GilCarvajal2024] demonstrates the value of motion and vibration, affordable integration of comprehensive multi-sensory feedback—including vestibular cues, seat vibrations mimicking road textures, wind simulation, and olfactory cues (e.g., engine smells)—remains technologically and economically challenging for driving schools.

Gap 12: Real-Time Physiological Monitoring for Safety Assessment. While EEG [Qadir2019], EDA [Muguro2023], and eye-tracking [Chung2022] show promise for cognitive load and attention monitoring, practical, non-invasive, real-time physiological feedback systems integrated into consumer VR headsets for continuous driver state assessment are not commercially available or validated for training environments.

2.1.5 Regulatory, Ethical, and Commercialization Gaps

Gap 13: Regulatory Approval and Certification Standards. There is minimal research on how VR-based training hours can be officially recognized by transport authorities (e.g., RTOs in India, DMVs in the US) as equivalent to on-road training hours. Establishing validation protocols, safety benchmarks, and certification standards for VR driving simulators in compliance with national/international regulations is an open challenge.

Gap 14: Intellectual Property and Patentability Analysis. Despite the commercial potential of VR driving training systems, few studies [Jurik2021, Silvera2022] discuss patent landscapes, prior art analysis, or strategies for protecting novel hardware configurations, software architectures, and training methodologies as intellectual property.

Gap 15: Scalability and Deployment in Developing Countries. Most VR driving research originates from technologically advanced regions (USA, Europe, South Korea, Japan). Limited attention has been given to adapting VR training systems for developing markets with constrained budgets, unreliable power infrastructure, limited technical support, and diverse traffic conditions (e.g., mixed traffic with two-wheelers, pedestrians, and livestock in Indian urban environments).

2.1.6 Summary of Research Gaps

The identified gaps highlight five overarching themes requiring urgent research attention:

1. **Cost-effective, high-fidelity hardware** with wireless connectivity and dual-mode transmission support.
2. **Larger, more diverse longitudinal studies** with validated real-world transfer metrics.
3. **Standardized, adaptive training frameworks** aligned with regulatory requirements.
4. **Enhanced realism through multi-sensory feedback** and photorealistic rendering.
5. **Regulatory certification pathways and IP protection strategies** for commercial deployment.

The DriveXR project directly addresses Gap 1 by demonstrating that a consumer-accessible racing wheel (rather than custom hardware) can deliver the interaction fidelity the literature identifies as critical, Gap 7 by identifying the absence of standardised assessment metrics as an area for future work rather than prematurely claiming a solved rubric, and Gap 10 partially, through a modelled vehicle interior and three distinct training environments. During development, the originally proposed wireless Bluetooth-enabled, dual-mode (automatic/manual) control approach was reassessed: a wireless BLE controller was found to face the same underlying platform-level incompatibility with standalone VR headset deployment as any USB HID device, and manual transmission simulation was deprioritised in favour of establishing a reliable automatic-transmission core experience given single-developer project resourcing. These pivots, and the resulting scope of the implemented system, are documented in full in the Methodology, System Analysis, and Results chapters that follow.

Chapter 3

System Analysis and Design

3.1 Overview

The **VR Driving Simulator** project aims to provide a high-fidelity training and evaluation environment for driving schools. The simulator models real-world driving dynamics using virtual reality and custom-built physical input devices (steering wheel, pedals, and gear shifter). This chapter presents a comprehensive analysis of requirements, constraints, design alternatives, and system architecture that directly address the research gaps and problem statement identified in Chapters ?? and ??.

3.2 Requirements Analysis

3.2.1 Functional Requirements

Based on the identified research gaps and the driving-training use case, the system was required to fulfill the following functional requirements:

FR1: Realistic Vehicle Control Input

- Accept analog steering input with proportional angle mapping across the full 900° rotation range of the connected racing wheel.
- Process independent accelerator and brake pedal inputs, correctly remapped for the connected hardware’s resting-position convention.
- Support gear selection for a four-position automatic transmission (Park, Reverse, Neutral, Drive), operable via keyboard, wheel buttons, and the physical H-pattern gear shifter.
- Maintain input responsiveness sufficient to preserve the driver’s motor-skill feedback loop, informed by the interaction-fidelity findings of Ebnali Harari et al. [**Harari2021**].

FR2: Immersive Visual and Audio Rendering

- Deliver stereoscopic 3D rendering at the Meta Quest 3’s native 72 Hz refresh rate to minimize motion sickness [Lindal2019, Weidner2017].
- Provide a field of view matching the Meta Quest 3’s native display specifications.
- Implement directional audio feedback including engine sound scaled to vehicle speed, horn, tyre skid, and collision impact sound [GilCarvajal2024].
- Render a modelled vehicle interior (Tata Safari dashboard, steering wheel, mirrors) and three distinct outdoor environments.

FR3: Vehicle Physics Simulation

- Model acceleration, braking, and cornering behaviour using Unity’s WheelCollider physics, with suspension and friction parameters tuned to approximate real-world handling.
- Implement speed-sensitive steering angle reduction to prevent unrealistic high-speed oversteer.
- Implement collision response via Unity’s physics engine and audio feedback scaled to impact velocity.

FR4: NPC Traffic and Environment Behaviour

- Populate the city environment with AI-controlled vehicles that navigate a checkpoint graph, respond to traffic signals, and yield appropriately at intersections.
- Provide three environments of progressive difficulty accessible via a VR-interactable menu, with instant switching and vehicle repositioning.
- Support VR hand-tracking interaction for in-cockpit controls (steering wheel grab, horn).

FR5: Preliminary Evaluation Support

- Support informal qualitative observation of driving behaviour across test sessions (Chapter ??).
- Provide a foundation for future structured, logged performance assessment — identified as a near-term extension rather than a requirement of the current implementation (Chapter ??).

3.2.2 Non-Functional Requirements

NFR1: Cost Effectiveness

- Use consumer-accessible hardware components — a Meta Quest 3 headset and a commercially available racing wheel and shifter — rather than custom-fabricated electronics, keeping total hardware cost well below dedicated commercial simulator installations [Silvera2022].
- Avoid recurring software licensing costs by using Unity’s free-tier licensing at the project’s current development scale.

NFR2: Usability

- Provide an in-VR tutorial overlay summarising controls at the start of a session, reducing the need for extensive verbal onboarding.
- Support both VR-headset operation and desktop/keyboard fallback for development and testing purposes.
- Provide a world-space VR menu for environment selection that responds to both hand-ray pointing and hand-tracking interaction.

NFR3: Reliability

- Ensure racing wheel and headset connections are established and verified before each session, as documented in the installation and user manuals.
- Design the NPC traffic controller with watchdog recovery logic so that vehicles which become stuck or blocked self-resolve rather than requiring manual scene intervention.

NFR4: Extensibility

- Structure the three training environments within a single Unity scene using a shared checkpoint-graph and traffic-light framework, allowing additional environments to be added following the same pattern.
- Keep vehicle physics, NPC AI, and VR interaction as separable components, allowing future extension (e.g., force feedback integration, additional vehicle models) without requiring a full architectural rewrite.

3.3 Constraints and Design Trade-offs

3.3.1 Hardware Constraints

C1: VR Headset Limitations

- The Meta Quest 3's mobile chipset limits achievable polygon count and texture resolution when rendering natively on-device.
- Trade-off: prioritise frame rate stability over maximum visual fidelity to reduce simulator sickness risk.
- Solution adopted: the system is deployed as a PC-hosted application streamed to the headset via Meta Quest Link, offloading rendering to desktop GPU hardware rather than the headset's onboard chipset. This resolves the constraint directly rather than requiring aggressive on-device optimisation.

C2: Racing Wheel Hardware Compatibility

- USB HID racing wheel hardware (Logitech G29) requires host-side driver initialisation that is not supported by the Quest 3's Android-based operating system in a standalone deployment.
- Trade-off: a fully standalone deployment would need to forgo physical wheel hardware in favour of hand-tracking-only or Quest controller input, sacrificing control realism.
- Solution adopted: the PC-hosted Meta Quest Link architecture (Section ??) allows the racing wheel to connect directly to the PC, where standard Windows USB HID drivers apply without limitation.

C3: Force Feedback

- Force feedback steering resistance, while supported by the G29's hardware, has not yet been integrated into the simulation software.
- Trade-off: reduced haptic realism for road-surface and cornering-load feedback versus development time allocated to core simulation stability.
- Mitigation: audio feedback (tyre skid sound, engine load) partially compensates; force feedback integration is identified as future work (Chapter ??).

3.3.2 Software and Performance Constraints

C4: Physics Engine Fidelity

- Unity's built-in WheelCollider physics component is optimised for general-purpose vehicle games rather than driver-training-grade accuracy.
- Trade-off: computational cost of high-fidelity tyre slip modelling versus frame rate stability.
- Solution adopted: WheelCollider suspension and friction curve parameters were iteratively tuned against qualitative real-world handling comparisons rather than implementing a full Pacejka tyre model, balancing accuracy against development time and runtime performance.

C5: NPC Traffic Behaviour Complexity

- Reliable multi-vehicle NPC navigation at signalised intersections is a recognised open problem in traffic simulation, not unique to this project.
- Trade-off: sophisticated traffic-flow algorithms (e.g., reinforcement-learning-based driver agents) versus a deterministic, debuggable rule-based approach appropriate for a training simulator where predictable AI behaviour is preferable to emergent complexity.
- Solution adopted: a finite state machine NPC controller (Chapter ??) with explicit states and yield-priority ordering, favouring predictability and debuggability over behavioural sophistication.

3.4 Design Alternatives and Justification

3.4.1 VR Platform Selection

Alternatives Considered:

1. **Valve Index + PC:** Pros: superior field of view (130°) and refresh rate (144 Hz). Cons: high combined cost, wired tethering by design, no native hand tracking.
2. **Meta Quest 3:** Pros: wireless standalone capability, affordable relative to PC-VR headsets, inside-out tracking, native hand tracking, and — critically for this project — Meta Quest Link support enabling a PC-hosted fallback architecture. Cons: lower on-device processing power than a dedicated PC-VR headset.
3. **PSVR2 + PlayStation 5:** Pros: affordable bundle cost, controller haptics. Cons: closed development ecosystem unsuitable for a research/education project requiring custom Unity development.

Selected: Meta Quest 3 — Justification: the Quest 3’s support for Meta Quest Link proved decisive, as it enabled the PC-hosted architecture that resolved the racing wheel compatibility constraint (C2) while retaining the headset’s native hand tracking, a requirement not met by pursuing a PC-VR-only headset that would need separate hand-tracking hardware.

3.4.2 Game Engine Selection

Alternatives Considered:

1. **Unity:** Pros: mature XR Interaction Toolkit with first-class Meta Quest Link and OpenXR support, extensive C# scripting ecosystem, large asset store for vehicle and environment assets.
2. **Unreal Engine:** Pros: superior default rendering fidelity, built-in Chaos vehicle physics. Cons: steeper learning curve for the development team, less mature Quest hand-tracking integration at project outset.
3. **CARLA Simulator:** Pros: purpose-built for driving simulation research. Cons: primarily Python-based with limited native VR headset support, requiring substantial custom integration work.

Selected: Unity 6 — Justification: Unity 6’s XR Interaction Toolkit provided direct, well-documented support for the Meta Quest Link and hand-tracking pipeline central to this project’s architecture, and the team’s existing C# familiarity favoured faster iteration during development.

3.4.3 Control Input Architecture

Alternatives Considered:

1. **Custom ESP32-based Bluetooth controls:** Pros: potentially lower per-unit hardware cost, wireless operation, full design customisability. Cons: requires custom PCB and sensor integration work, introduces calibration and reliability variables, and — most significantly — a Bluetooth Low Energy HID device still requires host-side driver support that the Quest 3 standalone environment does not provide, meaning this approach does not resolve the same underlying platform constraint (C2) as a commercial USB wheel connected via a PC-hosted architecture.
2. **Budget commercial racing wheel (Nitho Drive Pro One):** Pros: low cost, plug-and-play PC compatibility. Cons: found through direct testing to have insufficient rotation range (270° vs 900° for real vehicles), an undersized wheel, and pedal spacing too narrow for reliable use while wearing a VR headset (Chapter ??).
3. **Premium commercial racing wheel (Logitech G29):** Pros: 900° rotation range matching real vehicles, full-size wheel, force feedback capability, separated pedal set resolving the identification problem observed with the budget unit, and native USB HID compatibility with a PC-hosted architecture. Cons: higher unit cost than a custom or budget solution.

Selected: Logitech G29 via PC-hosted USB connection — Justification: direct user testing (Chapter ??) confirmed the G29 was rated substantially more realistic than the budget alternative, and its native USB HID compatibility with a Windows PC directly resolved the standalone-deployment driver incompatibility that motivated reconsideration of the originally proposed ESP32-based approach. This decision was made after the ESP32 approach was determined, during early development, to face the same platform-level input constraint as any USB or BLE HID device attempting standalone Quest 3 operation.

3.5 System Objectives

Based on the requirements analysis and design trade-offs, the system objectives are:

- To design a VR-based driving simulation environment replicating realistic vehicle control and handling behaviour.
- To integrate a commercially available racing wheel and pedal set providing physically realistic steering, throttle, and braking input.
- To ensure compatibility with the Meta Quest 3 headset via a PC-hosted Meta Quest Link architecture, retaining full head-tracking and hand-tracking functionality.
- To provide a structured, progressive-difficulty set of training environments suitable for novice driver familiarisation through to advanced intersection navigation.
- To develop a system using consumer-accessible hardware components, avoiding the cost and complexity of dedicated commercial simulator infrastructure.

3.6 Comparative Analysis with Existing Solutions

Table ?? compares DriveXR against existing commercial and research VR driving simulators based on key design parameters.

Table 3.1: Comparative analysis of VR driving simulator solutions

Parameter	DriveXR	DReyeVR [Silver et al. 2022]	Commercial Racing Sim	Traditional Simulator
Approx. Hardware Cost	Consumer VR headset + racing wheel	Consumer/research VR + PC	\$15,000–50,000	\$30,000–100,000
VR Platform	Meta Quest 3 (PC-hosted via Link)	Various HMDs (tethered)	Valve Index/Varjo (tethered)	Projector/screens (no VR)
Force Feedback	Hardware-capable, not yet integrated	Optional	Yes (high-fidelity)	Yes (hydraulic)
Hand Tracking	Yes (native Quest 3)	No	No	No
Control Hardware	Commercial racing wheel (Logitech G29)	USB wheel/keyboard	Proprietary	Wired industrial
Target Users	Novice drivers / driving schools	Researchers	Racing enthusiasts	Commercial training
Deployment	PC-hosted with VR headset	PC-VR only	PC-VR only	Fixed installation
NPC Traffic	Yes (state-machine controller)	Limited	N/A	N/A

Key Differentiators:

- **Accessible hardware:** built entirely from consumer-available components — a Meta Quest 3 headset and a commercial racing wheel — without requiring custom electronics fabrication.
- **Native hand tracking:** unlike DReyeVR or typical commercial simulators, DriveXR retains Quest 3 hand tracking for in-cockpit interactions (steering wheel grab, horn) alongside physical wheel input.
- **Structured progressive training:** three environments of increasing difficulty support a defined skill-building progression not present in research-focused platforms.

- **Integrated NPC traffic:** a purpose-built state-machine traffic controller for intersection navigation training, an area many lightweight research platforms do not prioritise.

3.7 System Components and Architecture

The system architecture consists of three tightly integrated subsystems.

3.7.1 Hardware Interface Subsystem

- **Racing Wheel:** Logitech G29 Driving Force, offering 900° steering rotation, force feedback capability, and separated accelerator/brake/clutch pedals connected via USB to the host PC.
- **Gear Shifter:** Logitech Driving Force Shifter, an H-pattern mechanical shifter connecting to the G29 base, providing physical gear selection input.
- **VR Headset:** Meta Quest 3, providing head tracking, optical hand tracking, and stereoscopic display, connected to the host PC via Meta Quest Link (wired USB-C or wireless Air Link).

Both the racing wheel/shifter assembly and the VR headset connect to the same host PC, which runs the Unity application and coordinates all input and rendering.

3.7.2 Software Simulation Subsystem

Developed in Unity 6 with the following modules:

- **Input Manager:** Reads racing wheel and pedal axis data via Unity’s Input System, applying deadzone filtering and the pedal-axis remapping required by the G29’s inverted resting-position convention (Chapter ??).
- **Vehicle Physics Engine:** Built on Unity WheelCollider components with custom-tuned suspension, friction, and drivetrain logic implementing front-wheel-drive behaviour and a four-position automatic gearbox.
- **Environment System:** Three hand-crafted training environments hosted within a single scene, with a state-machine-driven NPC traffic controller providing signalised intersection navigation in the city environment.
- **VR Interaction:** XR Interaction Toolkit providing head tracking, hand-tracking-based cockpit interaction (steering wheel, horn), and a world-space VR menu for environment selection.

3.7.3 VR Interface Subsystem

The Meta Quest 3 headset provides:

- **Head Tracking:** Six-degrees-of-freedom inside-out tracking via onboard cameras, passed through to the host PC via Meta Quest Link.
- **Hand Tracking:** Optical hand tracking used for in-cockpit interactions, similarly passed through via Meta Quest Link's OpenXR implementation.
- **Stereoscopic Rendering:** Rendered content is computed on the host PC GPU and streamed to the headset display via Meta Quest Link, rather than rendered natively on-device.

Deployment mode: PC-hosted via Meta Quest Link, connecting either by USB-C cable or wirelessly via Air Link.

3.8 System Architecture Diagram

Figure ?? illustrates the data flow and component interactions within the DriveXR system.

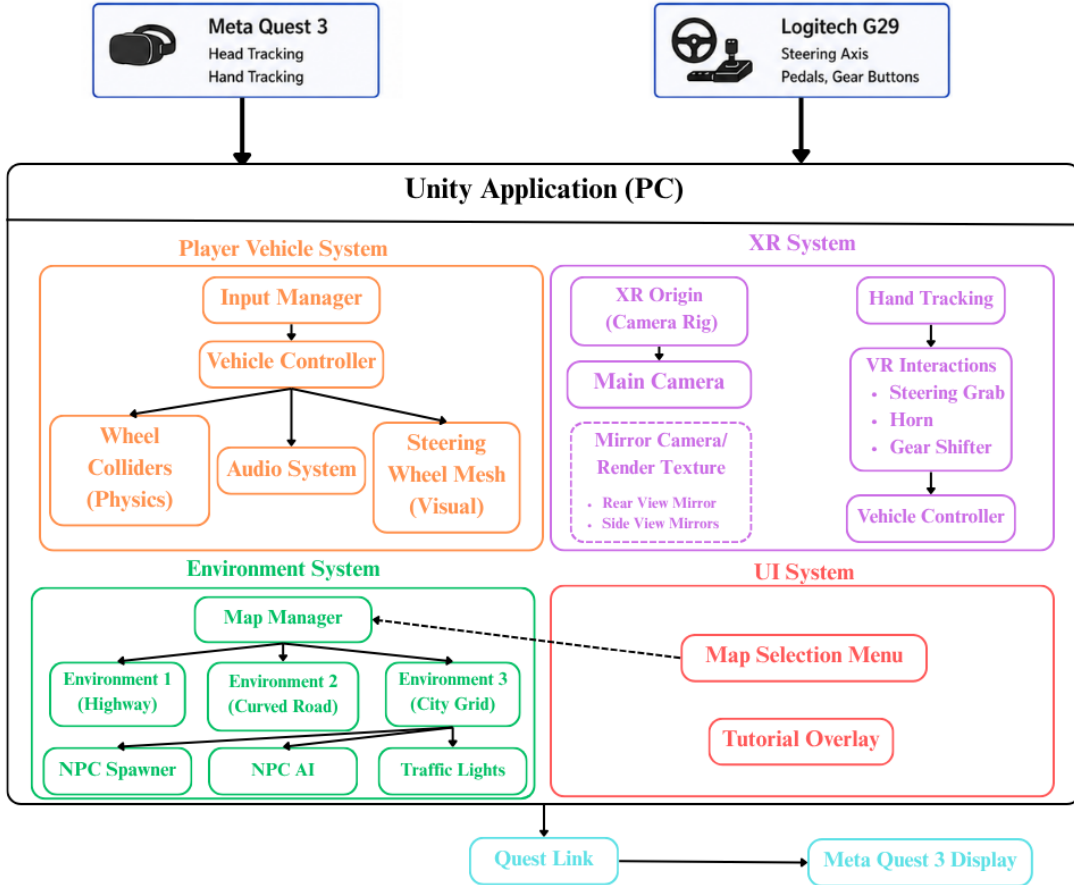


Figure 3.1: System architecture showing data flow from hardware inputs (racing wheel, headset tracking) through the Unity simulation engine to VR rendering output via Meta Quest Link.

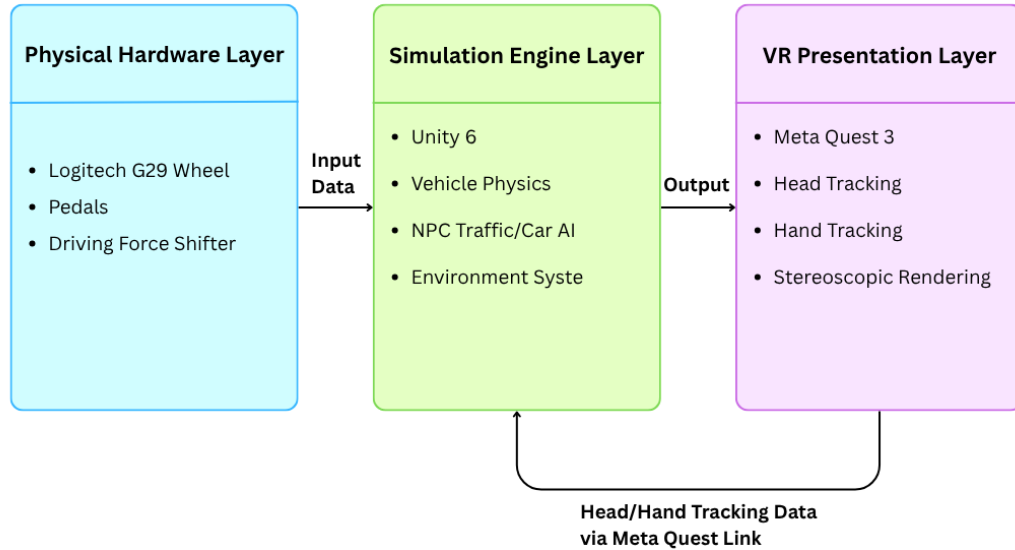


Figure 3.2: Conceptual system model illustrating the three principal layers of DriveXR — physical hardware input, the Unity-based simulation engine, and VR presentation via Meta Quest Link — and the data flow between them.

3.9 Design Rationale Summary

- **Consumer-hardware accessibility:** using a commercial racing wheel and consumer VR headset avoids the cost, fabrication risk, and calibration overhead of custom electronics while still meeting the project’s physical-realism requirements.
- **Platform compatibility resolution:** the PC-hosted Meta Quest Link architecture directly resolves the fundamental incompatibility between USB/BLE HID racing wheel hardware and standalone Quest 3 operation.
- **Retained VR capability:** despite not being a fully standalone deployment, the system retains full head-tracking and hand-tracking functionality identical to native operation.
- **Predictable NPC behaviour:** a deterministic finite-state-machine traffic controller was prioritised over more complex adaptive AI approaches, favouring debuggability and behavioural predictability appropriate for a training context.

3.10 Workflow and Operational Flow

The workflow diagram (Figure ??) shows the sequential process of system operation—from initialization through user interaction to post-session evaluation—illustrating how the system supports a complete training cycle.

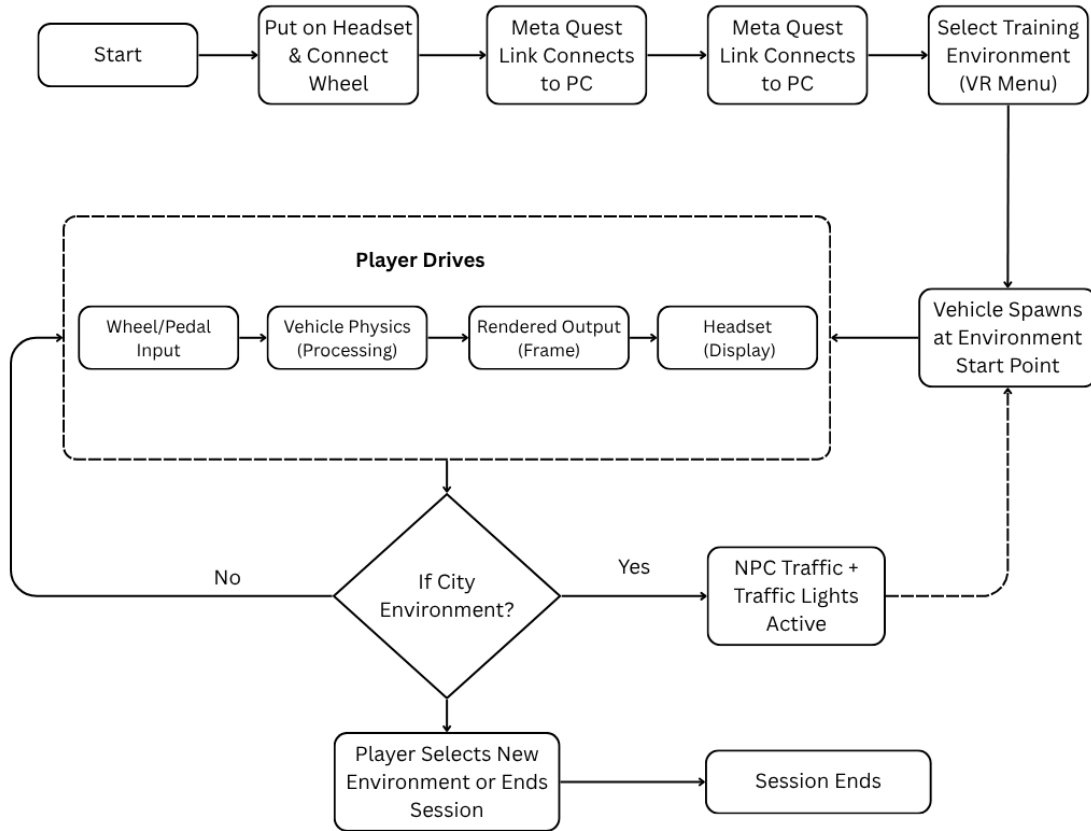


Figure 3.3: Operational workflow showing the complete training session lifecycle: pre-session setup, real-time simulation loop, and post-session assessment generation.

3.10.1 Detailed Functional Flow

1. Pre-Session Setup:

- The user connects the Logitech G29 wheel, pedals, and shifter to the host PC and starts Logitech G HUB.
- The user dons the Meta Quest 3 headset and connects it to the PC via Meta Quest Link (wired) or Air Link (wireless).
- The DriveXR application is launched on the PC and streams directly into the headset.
- An in-VR tutorial overlay appears, summarising the control scheme, and fades automatically or on user dismissal.

2. Environment Selection:

- The user opens the world-space VR menu via hand-ray pointing or the palm-up hand gesture.
- The user selects one of the three training environments (straight highway, curved road, or city grid).
- The vehicle’s position, rotation, and velocity are reset and it is placed at the selected environment’s spawn point.

3. Active Driving Loop:

- **Input Capture:** Unity’s Input System reads racing wheel steering and pedal axes each fixed update step; the gear shifter’s gate state is read via the Unity-Inputter package.
- **Physics Update:** Unity’s `FixedUpdate` applies motor torque, braking torque, and steering angle to the vehicle’s WheelCollider components, producing acceleration, cornering, and suspension response.
- **NPC Traffic (city environment only):** Each AI vehicle senses obstacles via a forward sphere-cast, checks its assigned traffic light state, and applies the appropriate speed and steering response (Chapter ??).
- **Audio:** Engine pitch and volume scale with vehicle speed; horn, tyre skid, and collision sounds trigger on their respective conditions.
- **VR Rendering:** The stereoscopic camera renders the driver’s viewpoint each frame, streamed to the headset via Meta Quest Link at the Quest 3’s native 72 Hz refresh rate.

4. VR Hand Interaction (concurrent with driving):

- The user may grab the in-VR steering wheel model with a tracked hand to steer via hand tracking in addition to the physical wheel.
- The user may press the in-VR horn button via hand tracking to sound the horn.

5. Session End:

- The user removes the headset and closes the application, or switches to a different environment via the VR menu to continue in a new session context.
- No automated scoring, PDF report generation, or persistent trainee profile exists in the current implementation; session observations were instead captured qualitatively by the observing researcher during testing (Chapter ??). Automated, logged performance assessment is identified as future work (Chapter ??).

3.11 Feasibility Study

This section evaluates the feasibility of the DriveXR project across technical, economic, operational, schedule, and legal dimensions, based on the project’s actual development experience and outcomes rather than solely prospective projection.

3.11.1 Technical Feasibility

Hardware Capabilities and Limitations: The project relies on a consumer-grade VR headset (Meta Quest 3) and a commercially available racing wheel and shifter (Logitech G29 and Driving Force Shifter) rather than custom-fabricated control hardware. Technical feasibility was demonstrated in practice through the following:

- **VR Headset Performance:** The Meta Quest 3 offers a native 72 Hz refresh rate and inside-out head and hand tracking, which proved sufficient for comfortable operation once the PC-hosted Meta Quest Link architecture was adopted (Chapter ??). Testing confirmed stable rendering at this native refresh rate across all three environments.
- **Racing Wheel Input Latency:** The Logitech G29 connects via USB directly to the host PC, and Unity’s Input System reads its axes and buttons without the latency and pairing concerns associated with a wireless BLE control scheme. This was one of the deciding factors, alongside the platform incompatibility discussed in Section ??, in moving away from the originally proposed custom wireless controller approach.
- **Physics Simulation Complexity:** Unity’s built-in WheelCollider physics component, running at Unity’s default fixed physics timestep, proved adequate for the vehicle handling characteristics targeted by this project once suspension and friction parameters were iteratively tuned (Chapter ??). No custom tyre-slip modelling beyond Unity’s built-in friction curves was required to achieve acceptable qualitative handling feel, as confirmed in testing (Chapter ??).
- **Integration Challenges:** The most significant technical integration challenge encountered was not cross-platform build configuration in the abstract, but the specific incompatibility between USB HID racing wheel hardware and a standalone Quest 3 deployment, which required the architectural pivot to a PC-hosted Meta Quest Link model documented in Chapter ?. A secondary integration challenge was reading the Logitech Driving Force Shifter’s gear gates, which required a third-party Unity package (Unity-Inputter) wrapping the Logitech G HID SDK, since the shifter’s button states were not reliably exposed through Unity’s legacy Input Manager on this system.

Risk Assessment: The technical risks that materialised during development were concentrated in NPC traffic AI reliability (vehicles becoming stuck or deadlocked at intersections) rather than in rendering performance, which remained stable throughout testing. The NPC controller was substantially reworked as an explicit state machine with watchdog recovery logic to address this (Chapter ??), though some limitations remain at high intersection traffic density (Chapter ??).

3.11.2 Economic Feasibility

Cost-Benefit Analysis: Economic viability is evaluated by comparing the actual development cost incurred against the cost of commercial alternatives, based on hardware acquired during this project rather than a hypothetical projected cost.

Development Costs Incurred:

- **Racing Wheel and Shifter:** Logitech G29 Driving Force and Logitech Driving Force Shifter, acquired via project funding request (Appendix ??).
- **VR Headset:** Meta Quest 3, used as the primary development and testing headset.
- **Software:** Unity 6 Personal edition (no licensing cost at current development scale); no paid third-party asset packages were required beyond the vehicle and environment models used.
- **Development Labour:** Single-student development effort over the current project timeline, supervised by the project guide.

Comparative Economics: Commercial dedicated driving simulators — including motion-platform and multi-screen installations used in some professional driving instruction contexts — typically cost several times more than the consumer hardware combination used in this project (a Meta Quest 3 headset and a mid-range commercial racing wheel). Precise return-on-investment or break-even projections for driving-school deployment are not presented here, as this project has not yet reached a pilot deployment stage; such analysis is identified as future work contingent on the formal user study and any subsequent institutional partnership (Chapter ??).

Economic Verdict: The hardware cost of the current system compares favourably against commercial simulator alternatives on a qualitative basis. A rigorous cost-benefit and ROI analysis requires deployment-scale data not yet available at this stage of the project and is deferred to future work.

3.11.3 Operational Feasibility

User Acceptance and Usability: Operational feasibility has been assessed qualitatively through informal testing sessions with the supervising faculty member and additional volunteer participants, as detailed in Chapter ??. No formal instructor onboarding protocol, curriculum integration study, or structured usability survey has yet been conducted; these are identified as necessary steps prior to any driving-school deployment.

- **Learner Comfort:** Preliminary observations (Chapter ??) indicate that simulator sickness is present in VR-inexperienced users initially, with symptoms reducing across repeated sessions. This is consistent with patterns reported in prior literature on VR acclimatisation, but has not yet been quantified for this specific system using a standardised instrument such as the Simulator Sickness Questionnaire.
- **Hardware Setup:** The system requires a PC capable of running Unity applications, a Meta Quest 3 headset, and the racing wheel/shifter assembly connected via USB — a moderate but not extensive infrastructure requirement compared to dedicated simulator installations.
- **Maintenance:** No formal durability or long-duration reliability testing has been conducted at this stage. Hardware maintenance considerations are limited to standard care of consumer VR and racing wheel peripherals.

Organisational Readiness: No formal engagement with driving schools or regulatory bodies has taken place at this stage of the project. Any future deployment consideration would require the structured evaluation and institutional partnership discussed in Chapter ??.

Operational Verdict: Provisionally feasible based on informal testing, but formal operational feasibility — instructor training requirements, curriculum integration, and institutional adoption — remains to be established through the structured evaluation planned as future work.

3.11.4 Schedule Feasibility

Project Timeline and Milestones: A six-month development schedule is proposed, aligned with academic project timelines and resource availability.

Table 3.2: Project Development Schedule

Phase	Status	Deliverables
Phase 1: Core Vehicle Physics	Completed	Tata Safari model with tuned Wheel-Collider physics, four-position automatic gearbox.
Phase 2: XR Rig and Hardware Input	Completed	XR camera rig, PC-hosted Meta Quest Link architecture, Logitech G29 wheel and pedal integration, gear shifter integration via third-party package.
Phase 3: VR Hand Interaction	Completed	Hand-tracking-based steering wheel and horn interaction.
Phase 4: NPC Traffic System	Completed, ongoing refinement	Finite-state-machine traffic controller with obstacle avoidance, traffic light compliance, and intersection yield logic.
Phase 5: Environment Construction	Completed	Three training environments (straight highway, curved road, signalised city grid) with VR environment-selection menu.
Phase 6: Preliminary Testing	Completed	Informal test sessions with supervising faculty and volunteer participants (Chapter ??).
Phase 7: Formal Evaluation	Planned (future work)	Structured user study using SSQ and SUS instruments across a larger participant cohort.

Development Approach: Unlike the phased timeline originally proposed, actual development followed an iterative approach in which architectural decisions were revised in direct response to concrete technical constraints encountered during implementation. The most significant such revision was the shift from a proposed standalone Quest 3 deployment with custom Bluetooth hardware to the implemented PC-hosted Meta Quest Link architecture with a commercial racing wheel, discussed in detail in Chapter ??. This pivot was necessitated by a platform-level incompatibility discovered during development rather than being anticipated in the original project plan.

Resource Availability: Development was carried out by a single M.Tech student under the supervision of the project guide, rather than the multi-person team structure originally envisioned. This resource constraint is reflected in the project’s current scope — a fully functional core simulator

with preliminary evaluation, rather than a fully validated, deployment-ready system with a formal user study already completed.

Schedule Risks Realised During Development:

- **Hardware platform incompatibility.** The most significant schedule impact came from discovering, partway through development, that USB HID racing wheel hardware could not be used with a standalone Quest 3 deployment. Resolving this required an architectural pivot rather than a simple bug fix, consuming development time not accounted for in an initial plan built around the original ESP32-based approach.
- **NPC traffic AI reliability.** The first rule-based NPC traffic implementation required substantial rework into a state-machine architecture after repeated reliability failures (Chapter ??), representing a second major unplanned development effort.
- **Single-developer capacity.** With one student rather than a multi-person team, feature scope was necessarily prioritised toward a stable core simulator over the fuller feature set (manual transmission, automated assessment, multi-vehicle support) originally envisioned.

Schedule Verdict: The project’s actual schedule diverged meaningfully from its original proposal, primarily due to the two architectural pivots described above. The current state represents a complete, working core simulator with preliminary evaluation; formal user study, additional vehicle models, and automated assessment remain as future work rather than completed deliverables, reflecting a realistic reprioritisation given single-developer resourcing.

3.11.5 Legal and Regulatory Feasibility

Intellectual Property and Licensing:

- **Software Licensing:** Developed under Unity’s free-tier license, appropriate for the project’s current academic, non-commercial scope.
- **Asset and Package Usage:** The vehicle model and environment assets used are subject to their respective source licenses. The project additionally depends on a third-party open-source Unity package (Unity-Inputter) for reading the racing wheel shifter’s input, used under its published license terms.
- **Patent Landscape:** No prior art search or patent filing has been conducted at this stage of the project, as it remains an academic prototype rather than a commercialisation candidate.

Safety Considerations:

- **VR Safety Guidelines:** Standard Meta Quest 3 health and safety guidance applies (age restrictions, motion sickness disclaimers, guardian boundary setup). Testers were informed of potential simulator sickness prior to sessions.

- **Hardware Safety:** During development, an attempt to apply custom force-feedback centering to the racing wheel produced an unexpectedly strong force that caused physical damage to the wheel’s mount. This was identified, the feature was disabled, and it is documented as a caution for any future force-feedback implementation work (Chapter ??). This incident underscores the importance of conservative, incremental testing when working with force-feedback-capable hardware.
- **Training Validity:** Indian regulations do not currently recognise VR training hours as equivalent to on-road training hours. DriveXR is positioned purely as a supplementary familiarisation tool at its current stage, not as a substitute for mandatory on-road instruction. Any future regulatory engagement would require the formal longitudinal validation this project has not yet undertaken [He2022].

Legal Verdict: No legal blockers apply to the project at its current academic prototype stage. Any future commercial or institutional deployment would require formal licensing review, liability consideration given the hardware safety incident noted above, and engagement with relevant regulatory bodies regarding training-hour recognition — all identified as beyond the current project’s scope.

3.11.6 Overall Feasibility Conclusion

Table 3.3: Feasibility Assessment Summary

Feasibility Dimension	Rating	Key Justification
Technical	Demonstrated	Core simulator fully functional; NPC traffic and force-feedback integration remain areas of ongoing refinement.
Economic	Favourable (qualitative)	Hardware cost compares favourably to commercial alternatives; rigorous ROI analysis deferred pending deployment-scale data.
Operational	Preliminary	Positive informal testing feedback; formal usability study and instructor-training protocol not yet conducted.
Schedule	Adapted	Original phased plan revised significantly due to two major architectural pivots; current scope reflects single-developer resourcing.
Legal	Adequate for current stage	No blockers for academic prototype; commercial deployment would require further legal and regulatory engagement.

The feasibility assessment reflects a project that has successfully delivered a working, demonstrable VR driving simulator built from consumer-accessible hardware, while several dimensions originally assessed prospectively — economic ROI, operational readiness, and regulatory pathway — remain to be validated through the formal evaluation and any future institutional engagement identified

as future work in Chapter ?? . The project’s technical core is sound and functioning; its readiness for real-world deployment beyond a research and demonstration context is not yet established and should not be overstated.

Chapter 4

Methodology

4.1 Overview

This chapter documents the development methodology followed in building DriveXR, including the architectural decisions made, the implementation phases actually carried out, the hardware evaluation process, and the testing procedures used to validate the system. Unlike a purely prospective plan, this chapter reflects the methodology as executed, including the pivots made in response to practical constraints encountered during development.

4.1.1 System Workflow

Figure ?? illustrates the end-to-end operational workflow of a DriveXR training session, from initial hardware connection through to session completion.

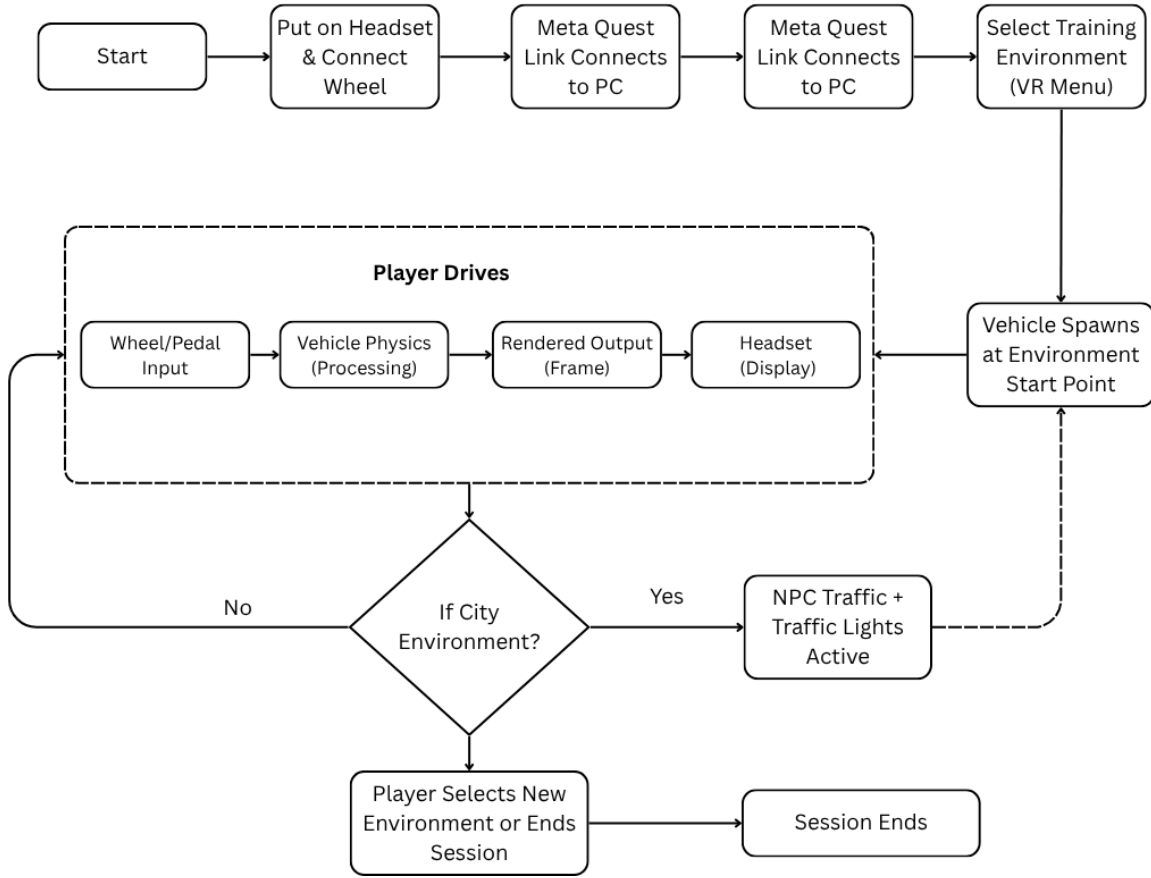


Figure 4.1: DriveXR operational workflow, from headset and hardware connection through environment selection, active driving, and session completion.

4.2 Architectural Decision: PC-Hosted Deployment via Meta Quest Link

An early architectural decision concerned the deployment target for the application. Two approaches were considered: a fully standalone build running natively on the Meta Quest 3’s Android-based operating system, or a PC-hosted build streamed to the headset via Meta Quest Link.

The standalone approach was initially pursued but presented a fundamental incompatibility: the Logitech racing wheel hardware communicates over a proprietary USB HID protocol that requires host-side driver initialisation. The Quest 3’s Android operating system does not support this initialisation sequence, and no accessible mechanism exists to sideload compatible drivers without device rooting, which would void hardware warranties and compromise the Meta Quest Link functionality required for hand tracking pass-through.

The PC-hosted architecture was adopted instead. In this configuration, the Unity application runs natively on a Windows PC. The Meta Quest 3 connects to the PC via USB-C or wirelessly via Air Link, and Meta Quest Link streams the rendered stereoscopic output to the headset while simultaneously relaying head-tracking pose data and hand-tracking skeletal data back to the PC

through the OpenXR runtime. The racing wheel, pedal set, and gear shifter connect directly to the PC via USB and are read natively by Unity’s Input System.

This architecture offers several practical advantages over the standalone approach:

- **Full hardware compatibility** — the racing wheel initialises normally as it would for any PC game, with no driver limitations.
- **Desktop-class rendering** — the application benefits from PC GPU rendering rather than being constrained to the Quest 3’s mobile chipset, permitting higher fidelity vehicle models and environment detail.
- **Retained VR capability** — head tracking and hand tracking function identically to a standalone deployment, since Meta Quest Link passes through the full OpenXR feature set including hand tracking subsystems.
- **Simplified build pipeline** — the Android build target introduces its own category of build issues (Gradle configuration, AAR dependency conflicts, manifest merging) that do not apply to a Windows Standalone build.

The trade-off is that a PC must be present during simulator operation, which is acceptable for the intended use case of supervised driving-school training sessions where an instructor workstation is expected to be present regardless.

4.3 Hardware Selection and Evaluation

4.3.1 Initial Hardware: Nitho Drive Pro One

Development began using a Nitho Drive Pro One, a budget-tier consumer racing wheel, to establish the core vehicle control pipeline. User testing during this phase revealed several limitations that motivated a hardware upgrade:

- Steering rotation limited to 270°, compared to the 900° rotation range of real vehicle steering.
- Undersized wheel diameter relative to an actual vehicle steering wheel, reducing physical immersion.
- No separate gear shifter peripheral.
- Accelerator and brake pedals positioned close together, causing testers — particularly those without prior VR experience — to misidentify which pedal they were pressing while wearing the headset, since the pedals are not visible during use.

4.3.2 Upgraded Hardware: Logitech G29 and Driving Force Shifter

Following an application for project funding (Appendix ??), a Logitech G29 Driving Force racing wheel and matching Logitech Driving Force Shifter were acquired. The G29 offers 900° rotation, a full-size wheel, force feedback, and substantially greater pedal spacing. All testers who evaluated both units rated the G29 as significantly more realistic. The increased pedal spacing was sufficient to resolve the pedal identification issue observed with the Nitho unit, without requiring a non-standard pedal remapping.

4.4 Development Phases

4.4.1 Phase 1: Core Vehicle Physics

The initial development phase established the vehicle controller using Unity’s WheelCollider physics components. A Tata Safari 3D model was rigged with four wheel colliders configured for front-wheel drive. Suspension parameters — spring strength, damper coefficients, and friction curves — were iteratively tuned to approximate the vehicle’s real-world ride characteristics. A four-position automatic gearbox (Park, Reverse, Neutral, Drive) was implemented with speed-gated gear-change restrictions to prevent unrealistic gear changes at speed.

4.4.2 Phase 2: XR Rig and Hardware Input Integration

The XR camera rig was anchored to a driver-seat transform inside the vehicle hierarchy so that head-tracked camera movement remained fixed relative to the cockpit rather than the physical room. Hardware input from the racing wheel was integrated through Unity’s legacy Input System, with a delayed detection routine to prevent the Quest 3 controllers — which also register as joystick devices when connected via Link — from being mistaken for the racing wheel.

A significant integration challenge arose from the G29’s pedal axis behaviour: the pedals report an axis value of +1 at rest and −1 at full travel, the inverse of the conventional 0-to-1 convention. This was resolved with the remapping function:

$$\text{input} = \text{Clamp}_{01}\left(\frac{-\text{raw} + 1}{2}\right)$$

4.4.3 Phase 3: VR Hand Interaction

VR hand tracking interaction was implemented for the steering wheel and horn using Unity’s XR Interaction Toolkit. A passive interaction component (rather than a physics-based grab interactor) was used on the steering wheel mesh to prevent the XR runtime from relocating the mesh transform during grab events — an issue encountered during early implementation where the grab interactable caused the wheel to orbit around the vehicle rather than rotating in place. The corrected implementation records the angular position of the hand relative to the wheel centre at the moment of grab,

frozen in the car body’s coordinate frame to prevent feedback from the wheel’s own rotation, and computes subsequent steering angle from the angular delta.

4.4.4 Phase 4: NPC Traffic System

An initial NPC traffic controller was implemented using per-checkpoint raycasting and a rule-based stop/go system. Testing revealed significant reliability problems: vehicles frequently entered head-on deadlocks at intersections, oversteered on sharp turns causing wheel-spinning-in-place behaviour, and in some cases collided with the player vehicle despite explicit avoidance logic.

The traffic controller was subsequently redesigned as an explicit finite state machine with five states: *Driving*, *Following*, *StoppedAtLight*, *CrossingIntersection*, and *Yielding*. Key design principles adopted in the redesign include: vehicles never reverse (eliminating a major source of erratic behaviour); the player vehicle is always treated as a yield-priority obstacle rather than an object to steer around; steering uses a single pure-pursuit calculation toward the next checkpoint, hard-clamped to a maximum angle to prevent in-place wheel spinning; and obstacle detection uses a compact three-ray forward sensor fan rather than multiple per-checkpoint raycasts.

4.4.5 Phase 5: Environment Construction

Three training environments were constructed within a single Unity scene to allow instant switching without scene reloading. A straight highway environment provides an obstacle-free space for control familiarisation. A curved road environment introduces sustained steering correction and speed management through a series of sharp bends. A city grid environment — structured as a square perimeter road bisected by one horizontal and one vertical internal road, forming nine intersections — provides signalised intersection navigation with AI traffic. A world-space VR menu, interactable via both hand-ray pointing and hand tracking, allows the player to select and teleport between environments.

4.5 Testing Procedures

4.5.1 Informal Iterative Testing

Throughout development, informal testing sessions were conducted with the supervising faculty member and additional volunteer participants. These sessions focused on identifying usability issues, hardware ergonomics problems, and simulator comfort concerns rather than collecting formalised quantitative metrics, given the project’s current stage of development.

4.5.2 Planned Formal Evaluation

Following completion of current refinements — height calibration, NPC traffic tuning, and full hand-tracking integration — a structured evaluation is planned incorporating the Simulator Sickness

Questionnaire (SSQ) and System Usability Scale (SUS) administered to a cohort of novice and experienced drivers, alongside quantitative performance logging of frame rate stability and NPC traffic incident frequency.

4.6 Summary

The development methodology for DriveXR followed an iterative approach in which architectural and hardware decisions were revised in response to concrete technical constraints encountered during implementation — most significantly, the shift from a standalone Quest 3 deployment to a PC-hosted Meta Quest Link architecture, and the shift from a budget racing wheel to the Logitech G29. This iterative process, documented in full in this chapter, produced a working simulator validated through informal testing, with formal evaluation identified as near-term future work.

Chapter 5

Implementation

5.1 Overview

This chapter presents the concrete implementation of DriveXR’s principal subsystems, illustrated with screenshots of the running application and representative code excerpts. Where Chapter ?? describes the development process and the decisions made, this chapter documents what was actually built, module by module.

5.2 Vehicle Physics Implementation

The player vehicle is a Tata Safari model driven by four Unity WheelCollider components configured for front-wheel drive. Figure ?? shows the in-VR cockpit view, including the steering wheel, dashboard instrumentation, and the environment visible through the windscreen.



Figure 5.1: In-VR cockpit view from the driver’s seat, showing the steering wheel, dashboard, and city environment.

Suspension parameters were tuned iteratively rather than derived analytically, since the priority was qualitative handling feel rather than strict physical accuracy:

```
void SetupSuspension(WheelCollider wc)
{
    JointSpring spring = wc.suspensionSpring;
    spring.spring = springStrength;
    spring.damper = springDamper;
    spring.targetPosition = 0.5f;
    wc.suspensionSpring = spring;
    wc.suspensionDistance = suspensionDistance;
    ...
}
```

The gearbox is a four-position automatic (Park, Reverse, Neutral, Drive). Gear changes to Park or Reverse are rejected above a configurable speed threshold, matching real vehicle behaviour and preventing unrealistic drivetrain damage in simulation.

5.3 Hardware Input Implementation

5.3.1 Racing Wheel and Pedals

The Logitech G29 connects to the host PC via USB and is read through Unity's legacy Input System. A defining implementation challenge was the G29's pedal axis convention: the pedals report +1 at rest and -1 at full travel, the reverse of the conventional 0-to-1 range expected by most vehicle controller implementations. This was corrected with the remapping:

```
brakeInput    = Mathf.Clamp01((-rawAccel + 1f) / 2f);
throttleInput = Mathf.Clamp01((-rawBrake  + 1f) / 2f);
```

5.3.2 Gear Shifter

The Logitech Driving Force Shifter presented a more significant implementation challenge than the wheel itself. Its gear gate signals were not reliably exposed through Unity's legacy Input Manager as either buttons or axes on this system configuration. This was resolved using the third-party Unity-Inputter package, which wraps the Logitech G HID SDK directly and exposes each gear gate as an indexable button state:

```
bool forward =
    g[LogitechG29.G29Button.Shifter1].isPressed ||
    g[LogitechG29.G29Button.Shifter2].isPressed ||
```

```
g[LogitechG29.G29Button.Shifter3].isPressed ||
...

```

Since DriveXR simulates a single-gear automatic transmission rather than a full manual gearbox, all forward gate positions on the physical H-pattern shifter map onto a single Drive gear, while one gate is mapped to Reverse. This preserves the tactile experience of a full-size shifter without requiring a manual-transmission simulation model.

5.4 VR and XR Interaction Implementation

5.4.1 Steering Wheel Hand Interaction

The in-VR steering wheel uses a passive interaction component (Unity’s `XRSimpleInteractable`) rather than a physics-based grab interactable. This design choice followed a specific implementation problem encountered early in development: using a grab-type interactor caused the XR runtime to reposition the wheel mesh’s transform directly, which combined with the mesh’s off-centre pivot point caused the wheel to orbit around the vehicle rather than rotate in place when grabbed. The corrected implementation instead calculates a normalised steering value from the hand’s angular position relative to the wheel centre, leaving the mesh’s own transform hierarchy untouched by the XR interaction system.

5.4.2 VR Environment Selection Menu

A world-space UI menu, built entirely in code rather than via the Unity UI designer, allows environment selection via both hand-ray pointing and direct hand-tracking touch interaction. The menu is anchored close to the user’s viewpoint by design, ensuring it is always readable and reachable regardless of where the player is looking when it is opened; as a result it is intentionally too close to the camera to be captured whole in a single screenshot, and is instead described here structurally. The menu presents three cards, one per training environment, each showing the environment’s name and a short description (e.g. “Beginner — get used to the controls” for the straight highway). Selecting a card triggers the teleport sequence described below.

Selecting an environment triggers a teleport operation that resets the vehicle’s Rigidbody velocity and angular velocity to zero before repositioning it, preventing any carry-over momentum from the previous environment:

```
rb.linearVelocity = Vector3.zero;
rb.angularVelocity = Vector3.zero;
playerCar.transform.SetPositionAndRotation(target.position, target.rotation);

```

5.5 NPC Traffic System Implementation

The NPC traffic controller was implemented as an explicit finite state machine after an earlier per-checkpoint raycast implementation proved unreliable (Chapter ??). Figure ?? shows the city environment with multiple AI vehicles navigating a signalised intersection.

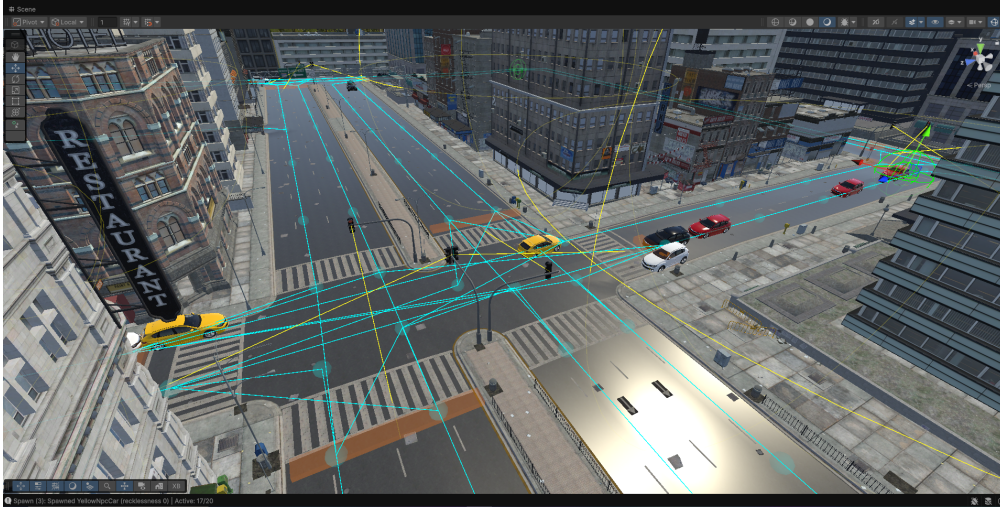


Figure 5.2: City environment showing NPC vehicles navigating a signalised intersection.

Obstacle detection uses a single forward sphere-cast originating from the vehicle's own centre, avoiding the sensor-offset issues present in the earlier implementation:

```
if (!Physics.SphereCast(origin, sensorRadius, transform.forward,
                        out RaycastHit hit, senseDist, seenLayers,
                        QueryTriggerInteraction.Ignore))
    return float.MaxValue;
```

A watchdog recovery mechanism snaps a vehicle back onto its path if it becomes stuck on geometry or blocked in traffic beyond a configurable timeout, ensuring that individual vehicle failures do not produce permanent gridlock:

```
if (tryingToMove && kmh < 2)
{
    stuckTimer += Time.fixedDeltaTime;
    if (stuckTimer >= stuckRecoverTime)
        Recover("stuck without progress");
}
```

5.6 Rear-View Mirror Implementation

Side and rear-view mirrors are implemented using dedicated off-screen cameras rendering into RenderTexture assets displayed on the mirror glass material. Each mirror camera is disabled by default and enabled for a single frame via a timed coroutine at 20 frames per second, reducing GPU overhead relative to continuous rendering. The mirrors are visible in the cockpit view already shown in Figure ??, displaying the environment behind and to the sides of the vehicle in real time.

5.7 Environment Implementation

All three training environments coexist within a single Unity scene with spatially separated road networks, allowing instant switching without a scene reload. Figure ?? shows a top-down Scene view screenshot of the complete layout, with the straight highway, curved road, and city grid all visible.



Figure 5.3: Top-down Scene view showing the complete layout of the three training environments within the single DriveXR scene: straight highway, curved road, and city grid.

5.8 Summary

This chapter has documented the concrete implementation of DriveXR’s vehicle physics, hardware input, VR interaction, NPC traffic, mirror rendering, and environment systems, with reference to the specific implementation challenges encountered and resolved during development. Chapter ?? presents the evaluation of these implemented systems.

Chapter 6

Testing

6.1 Overview

This chapter documents the functional testing carried out during DriveXR’s development. Testing was primarily iterative and issue-driven: features were implemented, tested interactively in the target environment (Unity Editor Play mode and, where applicable, the built PC application with the Meta Quest 3 connected via Meta Quest Link), and refined based on observed defects. This chapter presents the significant defects identified during this process as structured test cases, documenting the test performed, the expected behaviour, the actual (defective) behaviour observed, and the resolution applied.

6.2 Test Case Format

Each test case below follows a consistent structure: a unique identifier, the component under test, a description of the test performed, the expected outcome, the actual outcome observed, and the final status after resolution.

6.3 Vehicle Control Test Cases

Table 6.1: Vehicle Control Test Cases

ID	Component	Test / Expected	Actual Result / Resolution
TC-01	Gear shifting (Park brake release)	Shift from Park to Drive should release all brake torque and allow the vehicle to move.	Vehicle remained immobile after shifting out of Park; brake torque set in Park was never explicitly cleared on gear change. Resolved by adding an explicit brake-release pass at the start of each physics step whenever the gear is not Park and no braking input is active.
TC-02	Drive/Reverse direction	Selecting Drive should move the vehicle forward and Reverse should move it backward.	Directions were inverted — Drive moved the vehicle backward and Reverse moved it forward. Resolved by adding an <code>invertDriveDirection</code> toggle to the vehicle controller, allowing direction to be corrected without altering the core drivetrain logic.
TC-03	G29 pedal input mapping	Accelerator and brake pedals should map to 0 (unpressed) to 1 (fully pressed).	Vehicle accelerated immediately upon entering Drive with no pedal pressed. Root cause: G29 pedal axes rest at +1 and travel to -1, the reverse of the expected convention. Resolved with an explicit remap formula: <code>input = Clamp01((-raw + 1) / 2)</code> .
TC-04	Racing wheel device detection	The system should distinguish the physical racing wheel from Meta Quest controllers, both of which register as joystick devices when connected via Meta Quest Link.	Quest controllers were initially misidentified as the racing wheel, causing erratic uncommanded input. Resolved by filtering joystick device names containing "oculus", "quest", or "meta" during wheel detection.

Continued on next page

ID	Component	Test / Expected	Actual Result / Resolution
TC-05	Gear shifter input (H-pattern)	Moving the physical shifter through its gates should change the simulated gear accordingly.	No response — the shifter’s gate signals were not detected via Unity’s legacy Input Manager, either as axes or as standard joystick buttons, despite the hardware being confirmed functional via Logitech G HUB. Resolved by integrating a third-party Unity package (Unity-Inputter) that reads the shifter’s gate states directly via the Logitech G HID SDK.
TC-06	Force feedback centering spring	A moderate, constant centering force should be applied to the wheel to simulate steering resistance.	An initial implementation caused force to increase sharply with rotation angle, becoming strong enough to damage the physical wheel mount. Resolved by disabling the feature entirely pending safer, incremental parameter testing; documented as an open risk in Chapter ??.

6.4 VR and XR Interaction Test Cases

Table 6.2: VR and XR Interaction Test Cases

ID	Component	Test / Expected	Actual Result / Resolution
TC-07	Steering wheel hand grab	Grabbing the in-VR steering wheel with a tracked hand should rotate the wheel visually about its own centre.	The wheel instead orbited around the vehicle rather than spinning in place. Root cause: an XR grab interactable was repositioning the mesh transform directly, combined with an off-centre mesh pivot. Resolved by switching to a passive interaction component and calculating steering angle from hand position independently of the mesh's own transform.
TC-08	Steering wheel rest position	The steering wheel should remain at its designed rest orientation when the application starts.	The wheel snapped to a zero rotation at scene start rather than its intended tilted orientation. Resolved by capturing the wheel's actual rest rotation in <code>Awake()</code> and applying steering rotation as an offset from that captured value rather than overwriting it.
TC-09	VR menu button interaction	Pointing at and selecting a menu button via hand tracking should trigger the corresponding action.	Menu opened correctly via hand tracking, but button presses did not register. Resolved by confirming the Event System used an XR-compatible input module and configuring the canvas raycaster to support both ray and poke-based hand interaction.

Continued on next page

ID	Component	Test / Expected	Actual Result / Resolution
TC-10	Camera position stability	The VR camera should remain fixed relative to the driver's seat regardless of the player's physical position in their room-scale play area.	The Meta Quest 3's room-scale tracking allowed the camera to drift outside the vehicle mesh if the player physically shifted position. Resolved by implementing a camera-lock script that continuously corrects the XR Origin's position to maintain a fixed offset relative to the vehicle body, counteracting room-scale drift while preserving head rotation tracking.
TC-11	Hand tracking in PC-hosted build	Hand tracking should function identically whether running as a standalone Quest build or a PC-hosted build streamed via Meta Quest Link.	Hand tracking initially appeared not to function in the PC-hosted build. Resolved by confirming the Hand Tracking Subsystem feature was separately enabled under the PC tab of OpenXR project settings, distinct from the Android tab settings used for the earlier standalone build.

6.5 NPC Traffic Test Cases

Table 6.3: NPC Traffic System Test Cases

ID	Component	Test / Expected	Actual Result / Resolution
TC-12	Intersection light compliance	NPC vehicles already crossing an intersection when the light changes should complete their crossing rather than stopping mid-intersection.	Vehicles stopped abruptly in the middle of intersections when the signal changed during their crossing. Resolved by introducing an intersection-occupancy flag, set when a vehicle exits the stop-zone trigger, that overrides light-based stopping until the vehicle reaches its next checkpoint.
TC-13	Intersection dead-lock	Two NPC vehicles approaching an intersection head-on should not remain permanently blocked.	Vehicles would sometimes remain stopped indefinitely, facing each other with neither yielding. Resolved with a priority-ordering mechanism using a stable per-vehicle identifier, combined with a courtesy-yield timeout after which a blocked vehicle pauses briefly to allow the opposing vehicle through.
TC-14	Sharp turn navigation	NPC vehicles should complete turns at city intersections within a reasonable time.	Vehicles attempting sharp turns would sometimes rotate in place for extended periods (observed up to several minutes) without completing the turn, due to a steering angle limit narrower than the required turning radius. Resolved by implementing a speed-dependent steering limit (wider at low speed, narrower at high speed) and pre-planning reduced speed on the approach to sharp turns.

Continued on next page

ID	Component	Test / Expected	Actual Result / Resolution
TC-15	Player vehicle right-of-way	NPC vehicles should never collide with or attempt to steer around the player vehicle.	Some NPC vehicles occasionally collided with the player vehicle despite tag-based detection logic. Root cause: detection checked only the immediate collider's tag, missing cases where the player's child colliders were hit instead of the root object. Resolved by checking the collider, its parent hierarchy, and the transform root for the player tag and vehicle controller component.
TC-16	Vehicle stability during turns	NPC vehicles should not flip over while cornering.	Vehicles occasionally flipped, particularly during sharper turns. Resolved by lowering the vehicle's centre of mass, clamping angular velocity on the roll and pitch axes, and adding an active self-righting torque proportional to deviation from the vehicle's upright orientation.
TC-17	Traffic jam self-resolution	A blocked or stuck NPC vehicle should not cause a permanent, unrecoverable traffic jam.	Vehicles legitimately stopped behind an obstacle were not eligible for the stuck-recovery watchdog, since the watchdog only activated when a vehicle was attempting to move but failing, causing indefinite jams once one vehicle became stuck on geometry. Resolved by adding a second, independent blocked-timeout timer that triggers recovery for vehicles stopped behind an obstacle for an extended period, distinct from the geometry-stuck timer.

Continued on next page

ID	Component	Test / Expected	Actual Result / Resolution
TC-18	Off-road / fallen vehicle recovery	NPC vehicles should not become permanently lost if they fall through map geometry (e.g., at sidewalk collider gaps).	Vehicles occasionally fell through gaps between road and sidewalk colliders and became stuck below the visible road surface indefinitely. Resolved with a grounding watchdog that detects a significant drop below the vehicle's last known grounded height and snaps the vehicle back onto its path near its next checkpoint.

6.6 Summary

The test cases documented in this chapter reflect an iterative, defect-driven testing process characteristic of an actively developed system rather than a single formal test pass against a fixed specification. Several defects — particularly those relating to the NPC traffic controller and the racing wheel's gear shifter input — required substantial rework rather than isolated bug fixes, and are discussed in greater architectural detail in Chapter ?? . The resolution of the force feedback centering spring defect (TC-06) is highlighted specifically as a hardware safety consideration for any future work extending force feedback functionality.

Chapter 7

Results

7.1 Overview

This chapter presents preliminary evaluation results from testing conducted during DriveXR’s development. Results are drawn from informal test sessions involving the supervising faculty member and additional volunteer participants. As noted in Chapter ??, a formal structured user study using standardised instruments is planned as near-term future work; the observations presented here are qualitative and observational in nature.

7.2 Hardware Fidelity Evaluation

7.2.1 Racing Wheel Comparison

The Logitech G29 was evaluated directly against the Nitho Drive Pro One used during earlier development. All testers who used both units rated the G29 as substantially more realistic. The two most frequently cited factors were the G29’s 900° steering rotation, which matches real vehicle steering behaviour, and the larger physical wheel diameter, which more closely approximates an actual vehicle’s steering wheel.

7.2.2 Pedal Configuration

Pedal identification while wearing the headset was a specific concern raised early in testing. With the Nitho unit, testers — particularly those attempting the simulator for the first time — struggled to distinguish the accelerator from the brake pedal by feel alone, since neither pedal is visible while the headset is worn. An interim mitigation was tested in which the clutch pedal (positioned further from the accelerator) was remapped to serve as the brake, increasing the physical separation between the two pedals used for driving. Following the switch to the G29, whose pedal spacing is substantially greater than the Nitho’s, this remapping was found unnecessary and the standard accelerator/brake pedal assignment was restored without reintroducing the identification problem.

7.3 VR Comfort and Simulator Sickness

Simulator sickness was observed in several testers with no prior VR experience, most commonly manifesting as mild nausea during vehicle acceleration and turning manoeuvres. This is consistent with established findings that visual-vestibular mismatch is a primary driver of VR-induced discomfort in driving simulation contexts. Notably, symptom severity reduced substantially across repeated sessions — testers who reported discomfort in their first session generally reported markedly reduced or absent symptoms by their second or third session, suggesting an adaptation effect consistent with prior literature on VR acclimatisation.

The in-cockpit view was rated as comfortable during normal driving by all testers. Two related issues were identified concerning the VR camera rig’s interaction with the Meta Quest 3’s room-scale tracking system:

1. **Height calibration.** The Quest 3 establishes its tracking origin based on the physical floor position at headset startup. Testers of different physical heights therefore experienced the virtual driver seat at different vertical positions relative to the physical cockpit geometry, with taller participants generally reporting that the seat position felt low relative to the steering wheel.
2. **Positional drift.** Because the Quest 3 uses room-scale positional tracking, testers who physically shifted position while seated could cause the tracked camera origin to move outside the bounds of the vehicle mesh, producing a view from outside or above the car. This was addressed during development with a camera-locking mechanism that ties the XR camera’s position to a fixed offset relative to the vehicle body rather than the physical room, eliminating positional drift while preserving head rotation tracking.

7.4 NPC Traffic System Evaluation

The redesigned NPC traffic controller (Section ??) was evaluated qualitatively following its implementation. Vehicle navigation performed reliably on straight road segments, with vehicles consistently reaching checkpoints and maintaining appropriate following distances from the vehicle ahead.

Sharp turns at city intersections remained an area of weaker performance: the steering angle clamp necessary to prevent wheel-spinning-in-place behaviour occasionally proved insufficient for tight intersection turning radii, resulting in vehicles temporarily struggling to complete turns. Intersection deadlocks, while substantially reduced from the earlier implementation through the introduction of explicit yield-priority logic based on vehicle instance ordering, were not fully eliminated at high vehicle density, with simultaneous arrivals at opposing intersection approaches occasionally producing extended mutual blocking. These findings are consistent with the broader literature identifying dense urban NPC traffic simulation as an open research problem, and are treated in this report as an identified limitation for continued refinement rather than a solved component of the system.

7.5 System Performance

The PC-hosted architecture via Meta Quest Link achieved stable rendering at the Meta Quest 3’s native 72 Hz refresh rate across all three training environments during test sessions, with no observed frame rate degradation with up to ten concurrent NPC vehicles active simultaneously in the city environment. The rear-view and side-mirror rendering system, which uses dedicated off-screen cameras throttled to 20 updates per second to manage GPU load, operated without perceptible latency to testers.

7.6 Summary

Preliminary testing confirms several of the project’s underlying assumptions: that physical hardware fidelity — particularly steering rotation range and pedal spacing — has a measurable effect on perceived realism, and that VR-based driving simulation is broadly comfortable for use once initial simulator sickness adaptation has occurred. Testing also surfaced concrete engineering challenges requiring further attention, most significantly headset height calibration for users of varying stature and NPC traffic behaviour at high intersection density. These findings directly inform the future work items discussed in Chapter ??.

Chapter 8

Conclusion and Future Work

8.1 Summary

This project set out to address well-documented limitations in traditional driver training: cost, safety risk, and limited access to higher-end vehicles during instruction. Through literature review and iterative development, DriveXR was designed and implemented as a VR-based driving simulator combining a consumer-grade Meta Quest 3 headset with a Logitech G29 racing wheel, running as a PC-hosted application via Meta Quest Link.

The system requirements analysis (Chapter ??) established the functional and non-functional requirements grounded in the reviewed literature. A key architectural decision made during development — and one that diverged from the originally proposed standalone-with-custom-hardware approach — was the adoption of a PC-hosted deployment model using commodity racing hardware rather than custom Bluetooth-enabled controls. This decision was driven by a concrete technical constraint: the Quest 3’s Android operating system does not support the driver initialisation sequence required by USB HID racing wheels, making a standalone deployment with physical wheel hardware infeasible without device modification. The PC-hosted architecture, detailed in Chapter ??, resolved this constraint while preserving full head-tracking and hand-tracking functionality via Meta Quest Link’s OpenXR pass-through.

Similarly, the originally proposed custom ESP32-based Bluetooth steering and pedal hardware was superseded by the Logitech G29, a commercially available racing wheel offering 900° rotation and force feedback, following an evaluation that found a budget consumer wheel (Nitho Drive Pro One) insufficiently realistic for the project’s aims. This substitution reduced development risk associated with custom hardware fabrication while still meeting the project’s core requirement of physically realistic control input, as confirmed through user testing (Chapter ??).

8.2 Contributions

This work makes the following contributions:

1. **A working PC-hosted VR driving simulator** combining consumer VR hardware (Meta Quest 3) with commodity racing peripherals (Logitech G29) via a Meta Quest Link streaming architecture, achieving desktop-class rendering quality without requiring dedicated simulation hardware or custom electronics.
2. **A three-environment progressive difficulty training framework** — straight highway, curved road, and signalised city grid — implemented within a single Unity scene with instant environment switching via a VR-interactable world-space menu.
3. **An adaptive NPC traffic controller** redesigned as an explicit finite state machine incorporating obstacle avoidance, traffic signal compliance, intersection yield-priority resolution, and player-vehicle right-of-way, developed iteratively in response to reliability problems identified in an earlier raycast-based implementation.
4. **Empirical hardware evaluation** comparing budget and premium racing wheel hardware, documenting the specific ergonomic factors — pedal spacing, wheel rotation range, wheel diameter — that affect usability for VR-headset-wearing users who cannot visually reference physical controls during use.
5. **Preliminary usability findings** on VR-specific challenges in driving simulation, including simulator sickness adaptation patterns across repeated sessions and headset height/positional calibration requirements arising from Quest 3 room-scale tracking in a fixed-seat context.

8.3 Limitations

Several limitations of the current system are acknowledged:

- **Validation is preliminary.** Testing to date has been informal, involving the supervising faculty member and a small number of additional participants. A structured user study using standardised instruments (SSQ, SUS) with a larger and more diverse participant cohort, as outlined in Chapter ??, has not yet been conducted.
- **NPC traffic reliability at high density.** While substantially improved through the state-machine redesign, intersection deadlocks and turning difficulties at sharp intersection geometries were not fully eliminated, particularly under high simulated traffic density (Chapter ??).
- **PC dependency.** Unlike a fully standalone deployment, the current architecture requires a connected PC, which — while acceptable for the supervised training context this project targets — limits portability compared to a self-contained headset-only solution.
- **No force feedback.** The G29 supports force feedback steering resistance, but this feature has not yet been integrated into the simulation, meaning road-surface and steering-load feedback are not currently simulated.

- **Height calibration is manual.** The current system does not provide an in-VR mechanism for users to adjust seated eye height, requiring scene-level configuration rather than a runtime calibration step.

8.4 Future Work

8.4.1 Near-Term Improvements

- **Seated-mode height calibration.** Implement a runtime calibration step allowing users to set their eye-level offset within the cockpit independent of the Quest 3's room-scale floor reference, directly addressing the height-related usability issue identified in testing.
- **NPC traffic refinement.** Widen intersection turning geometry and reduce simulated vehicle spawn density at high-traffic intersections to further reduce deadlock frequency, alongside continued tuning of the state-machine yield-priority thresholds.
- **Completion of hand-tracking integration.** Extend current hand-tracking interactions (steering wheel, horn) to cover the full control surface including gear shifter interaction and in-VR menu navigation without reliance on ray-pointing alone.
- **Formal user study.** Conduct the structured evaluation protocol outlined in Chapter ??, incorporating SSQ and SUS instruments across a larger participant cohort spanning novice and experienced drivers.

8.4.2 Medium-Term Extensions

- **Force feedback integration.** Implement steering resistance and road-surface feedback using the G29's native force feedback motor, informed by prior research linking haptic fidelity to training transfer effectiveness.
- **Additional training scenarios.** Extend the environment library to include adverse weather conditions, night driving, and emergency response scenarios such as sudden obstacle avoidance.
- **Performance assessment module.** Introduce structured, logged performance metrics — speed compliance, lane discipline, signal compliance, following distance — to support objective learner assessment, an area intentionally deferred during the current development phase to prioritise core simulation stability.
- **Standalone deployment investigation.** Explore alternative approaches to standalone Quest 3 deployment with physical hardware input, such as a PC-bridge companion application relaying wheel input to the headset over a local network connection, to reduce PC dependency while retaining hardware realism.

8.5 Final Remarks

DriveXR demonstrates that a credible VR driving training platform can be built from consumer-grade hardware — a Meta Quest 3 headset and a commercially available racing wheel — without requiring custom electronics or dedicated simulation infrastructure. The development process surfaced concrete engineering challenges not fully anticipated at the project’s outset, most significantly the platform incompatibility between USB racing wheel hardware and standalone Quest deployment, and the behavioural instability of early rule-based NPC traffic logic. Resolving these challenges — through the PC-hosted Meta Quest Link architecture and the state-machine traffic controller redesign respectively — constituted a substantial portion of the practical engineering contribution of this work.

The system in its current state provides a functional, testable foundation for VR-based driver training. The identified limitations, particularly around formal validation and NPC traffic behaviour at high density, define a clear and achievable roadmap for continued development toward a system suitable for structured evaluation in an educational deployment context.